

Electrical Engineering Reference — Figures Companion

Editio Unica

Stephen B. Johnson

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Chapter 1

EE-Book Figures Companion

This companion document collects supplementary figures for the *Electrical Engineering Reference* by topic. Unlike the figures embedded directly in the chapters, entries here are additional visualizations — generated from the scripts/ marimo notebooks, or photographs and diagrams sourced from public-domain and CC0 archives — that illustrate chapter concepts without being woven into the main narrative. It is built and distributed separately from the main reference and the problem sets.

Each entry follows the same figure format used in the chapters: a numbered caption and the image. Externally sourced figures include a Source: line citing the origin and copyright status; no attribution is legally required for public-domain/CC0 works, but the source is noted for verifiability.

Chapter 1: Power Engineering

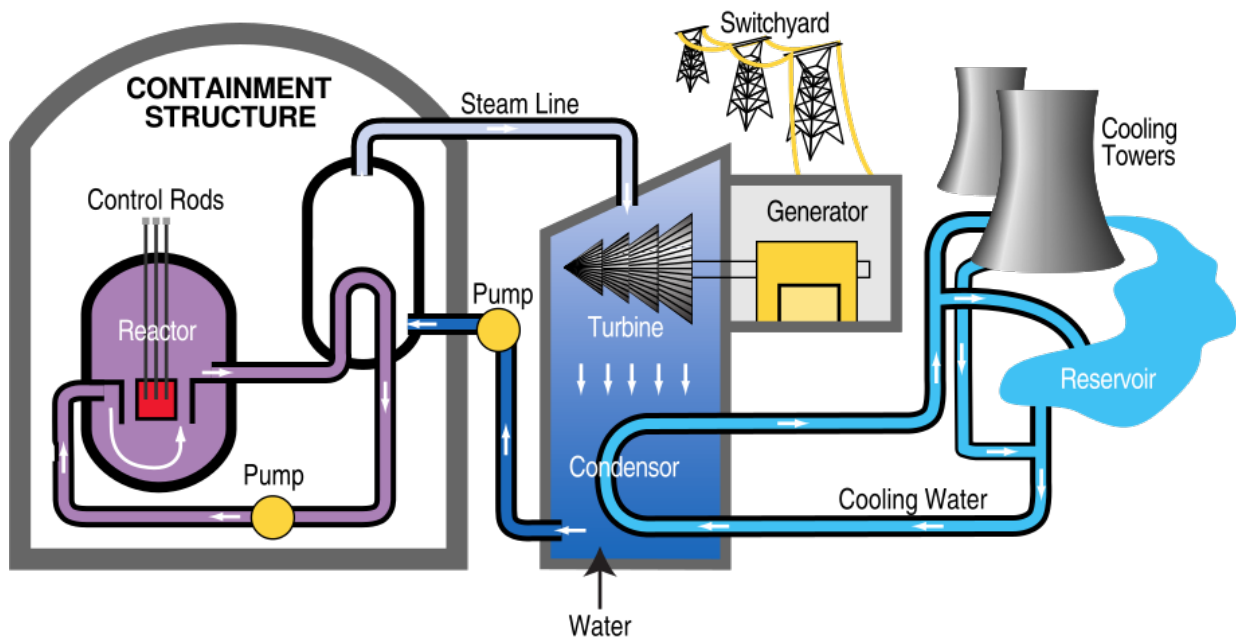


Figure C1.1: Pressurized Water Reactor (PWR) Nuclear Power Plant Diagram



Figure C1.2: High-Voltage Transmission Tower



Figure C1.3: Outdoor Power Distribution Substation



Figure C1.4: Substation Power Circuit Breaker, Metal-Clad Switchgear

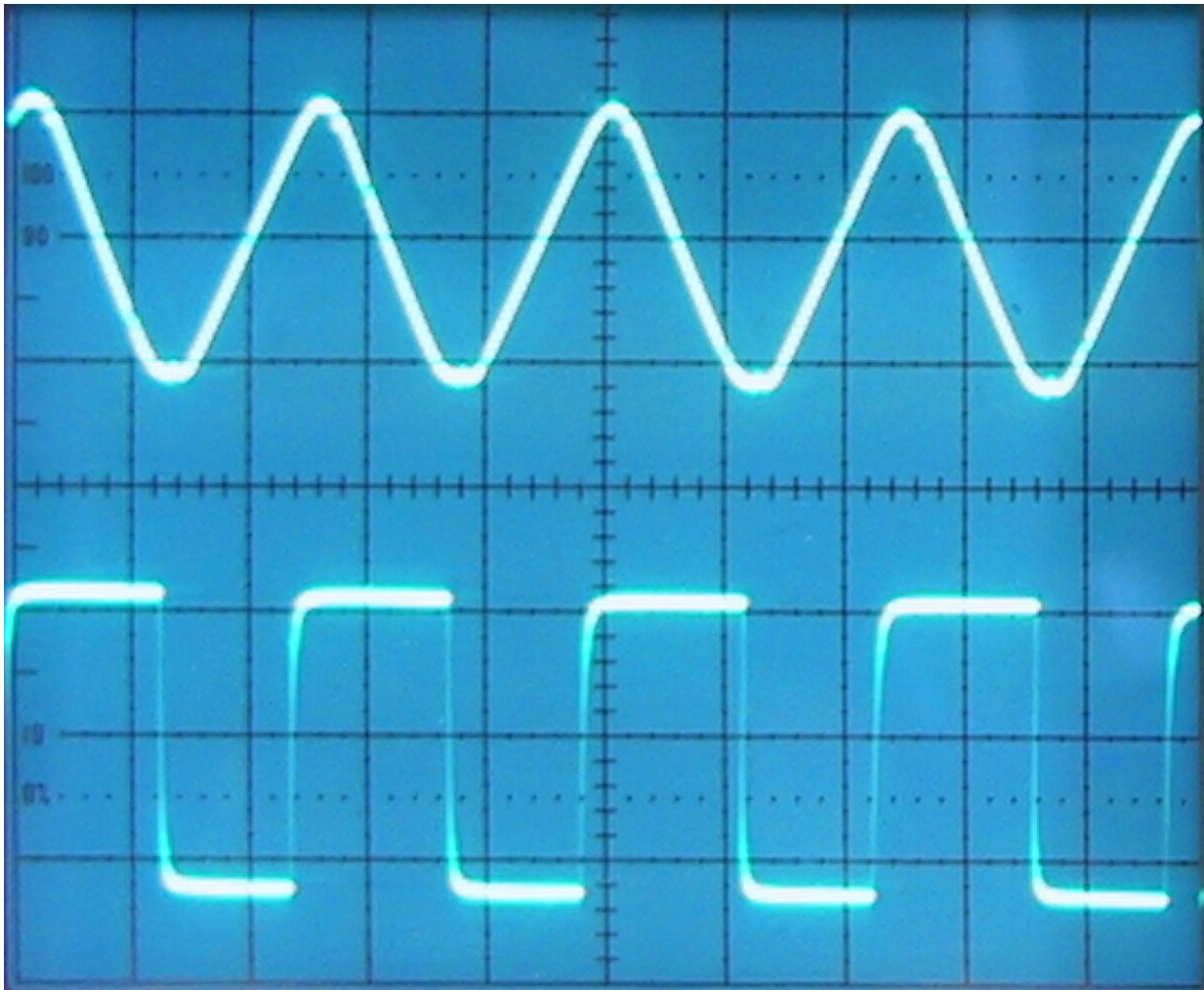


Figure C1.5: Fundamental and Harmonic Waveforms on an Oscilloscope

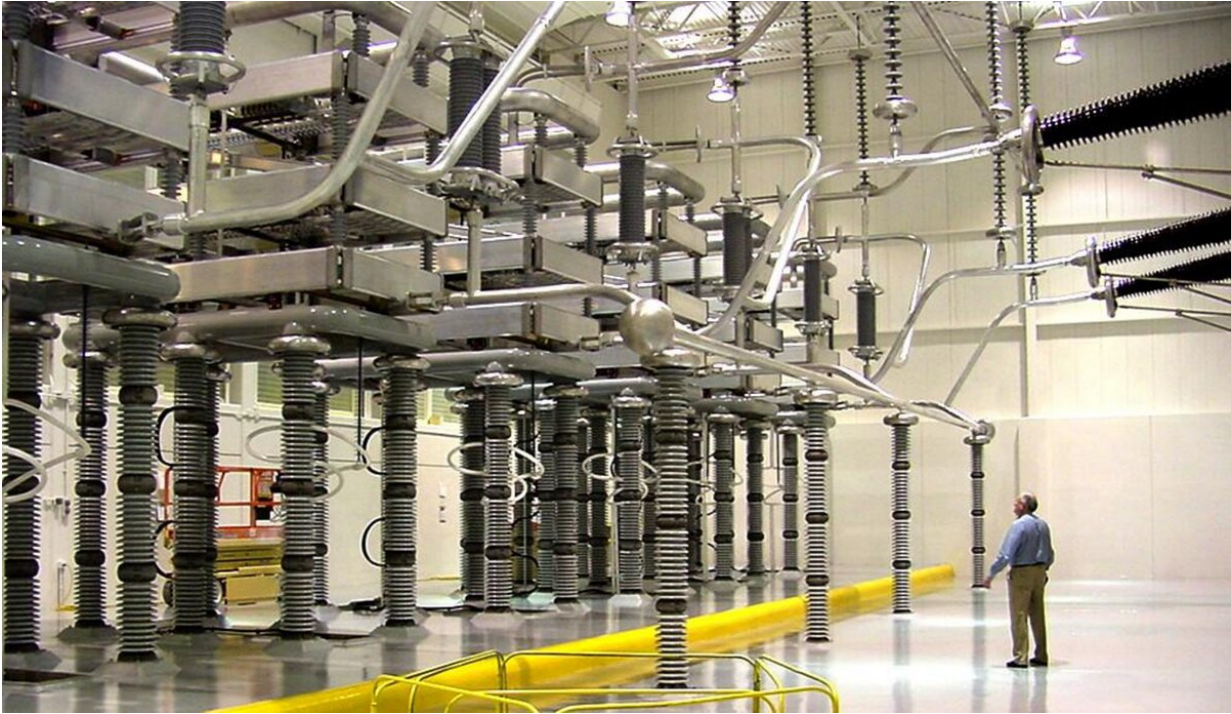


Figure C1.6: Celilo Converter Station, Pacific DC Intertie (HVDC Terminal)

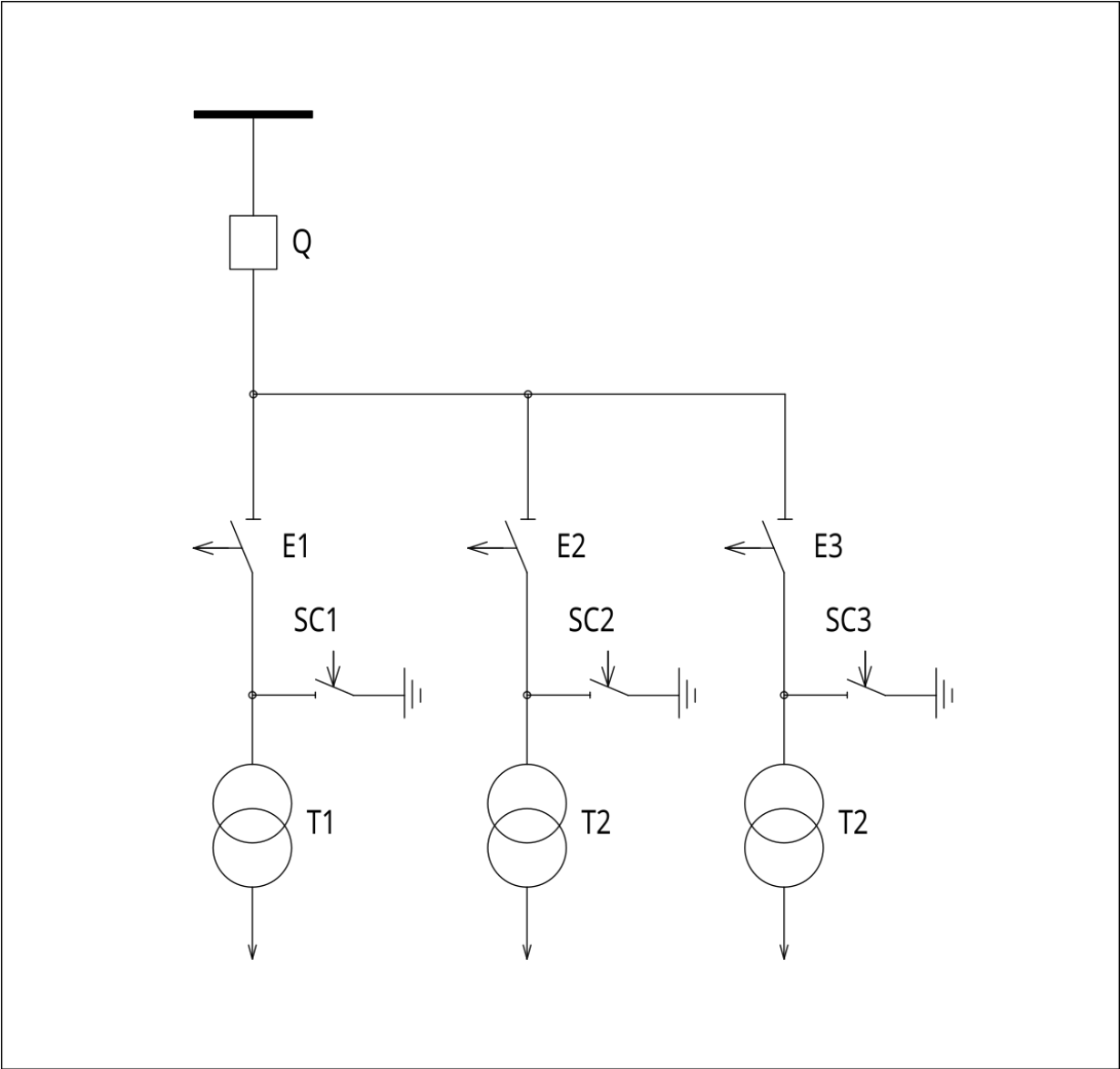


Figure C1.7: One-Line Diagram of a Distribution Substation



Figure C1.8: SCADA Operator Balancing Electrical Load

Chapter 2: Communications Engineering

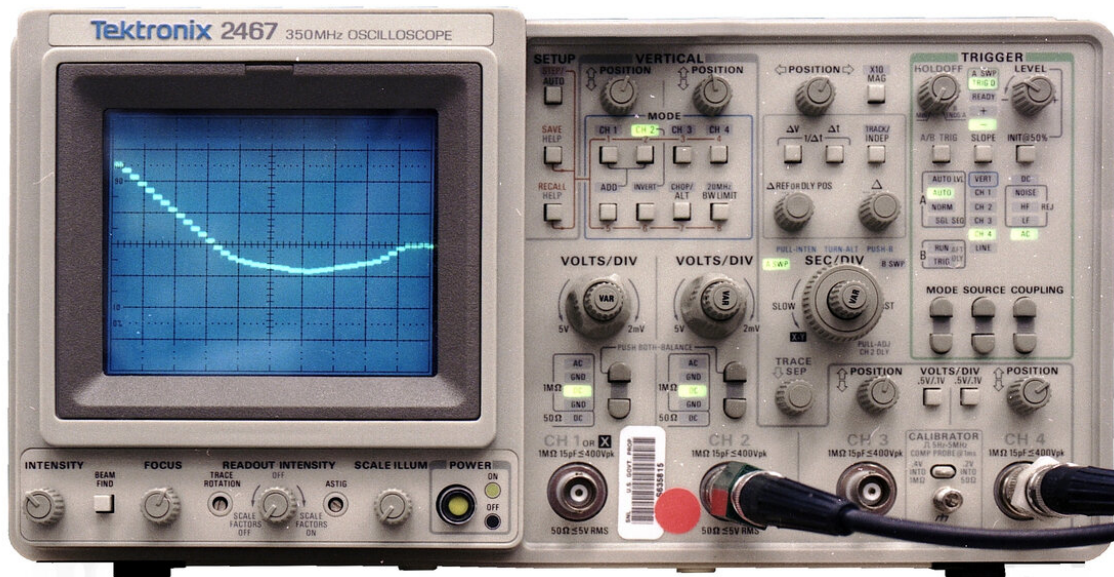


Figure C2.1: Analog Oscilloscope for Analog Signal Observation

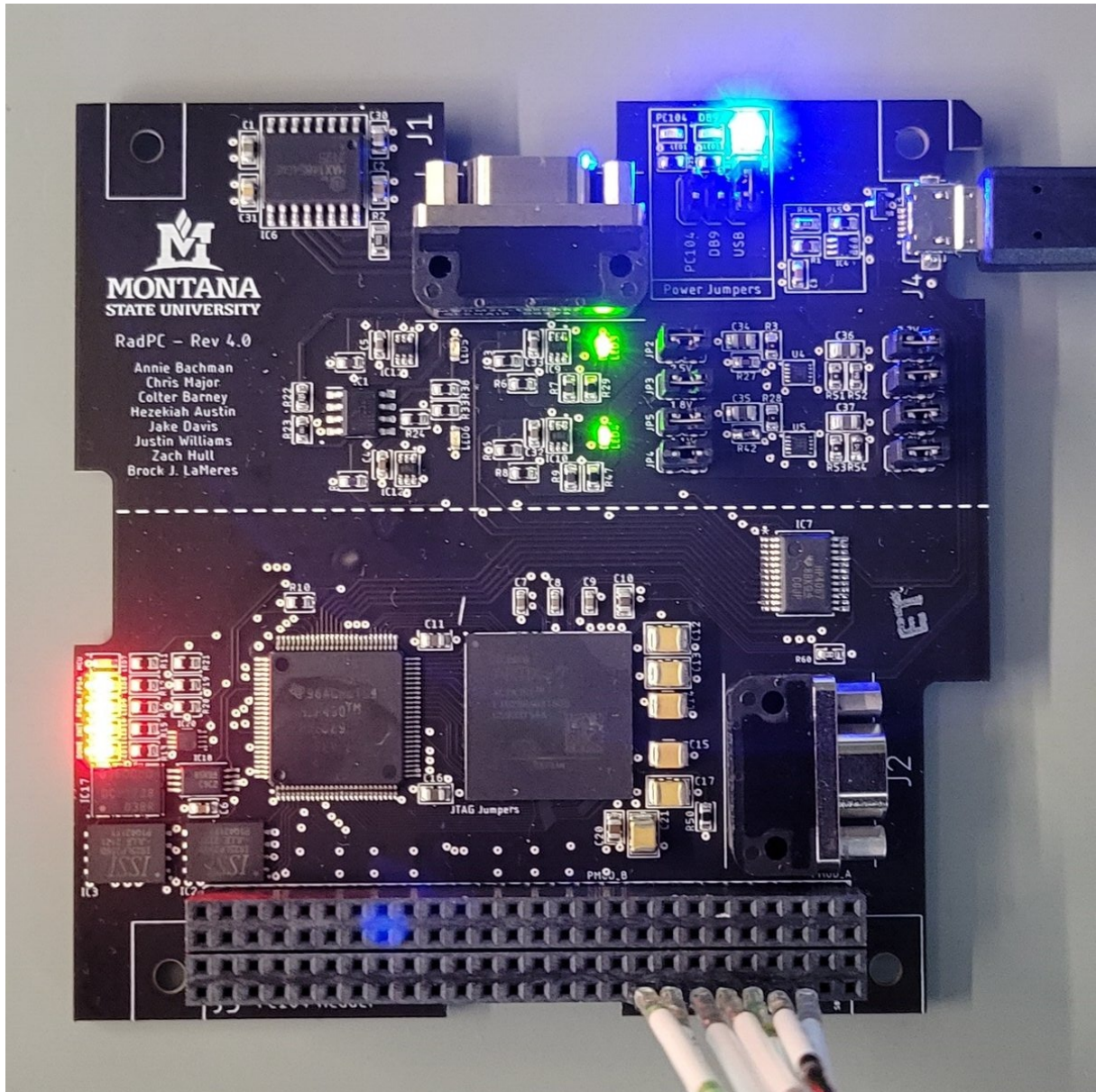


Figure C2.2: FPGA-Based Digital Signal Processing Board



Figure C2.3: HackRF One Software-Defined Radio Transceiver

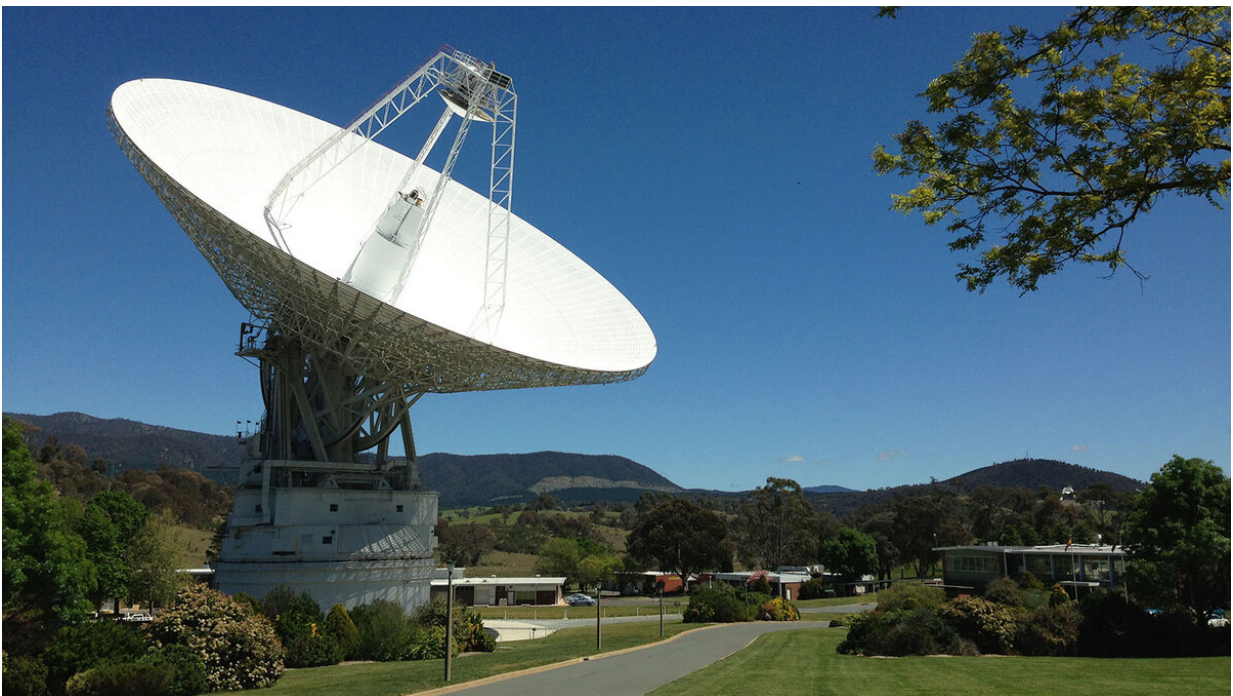


Figure C2.4: DSN Deep-Space Antenna (Canberra) — Channel Coding Heritage from Deep-Space Links



Figure C2.5: Manual Telephone Switchboard, Salt Lake City, 1907

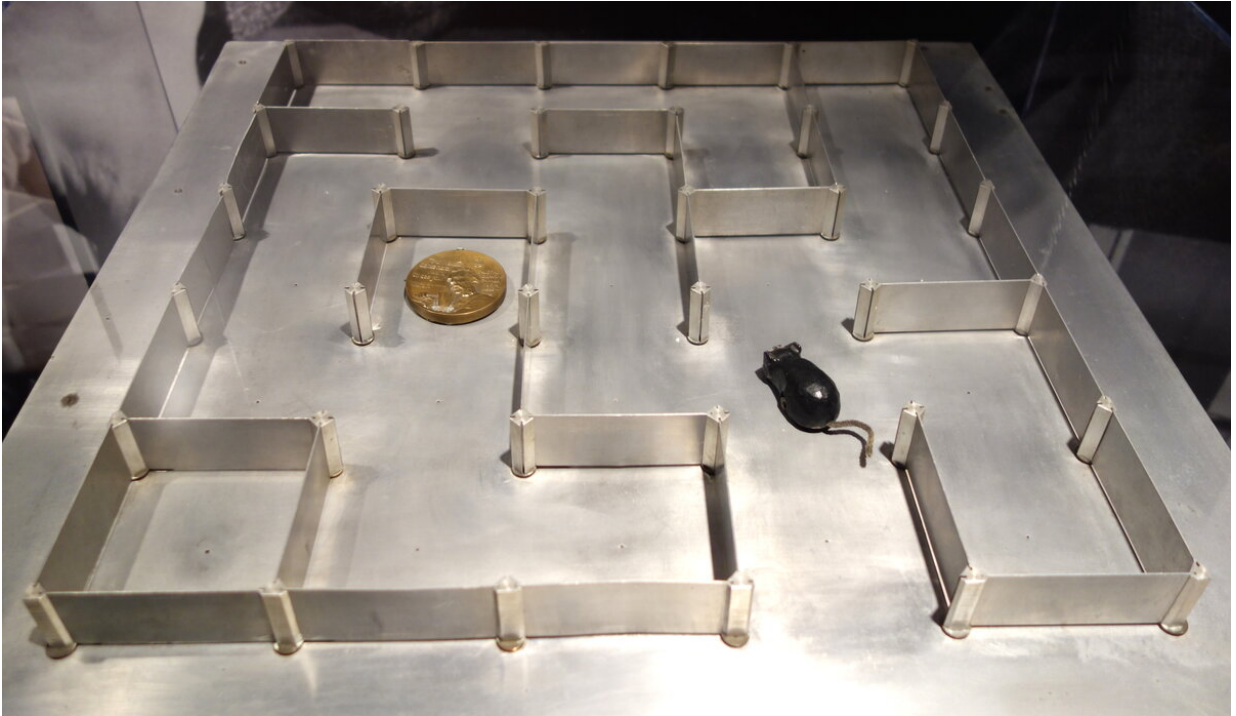


Figure C2.6: Theseus, Claude Shannon's Maze-Solving Mouse (1952)

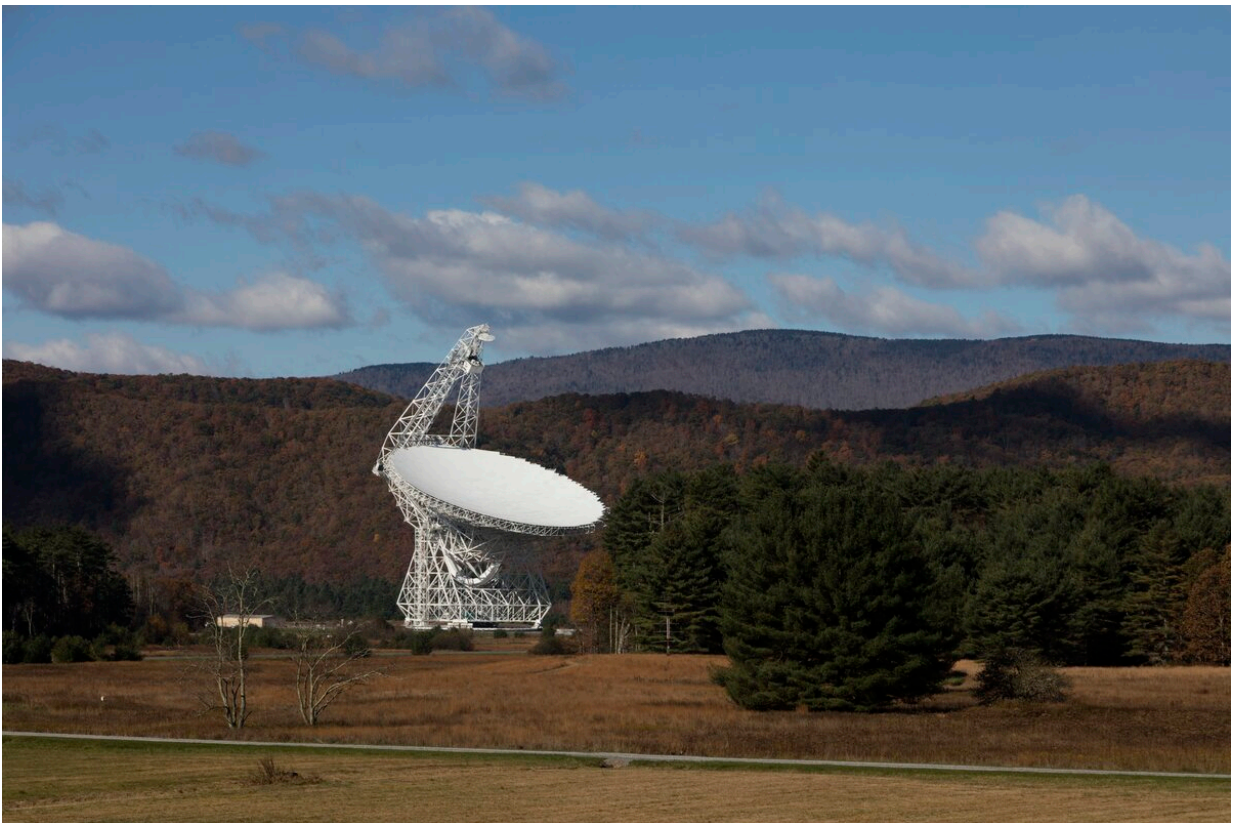


Figure C2.7: Green Bank Radio Telescope — Low-Noise Receiver Systems

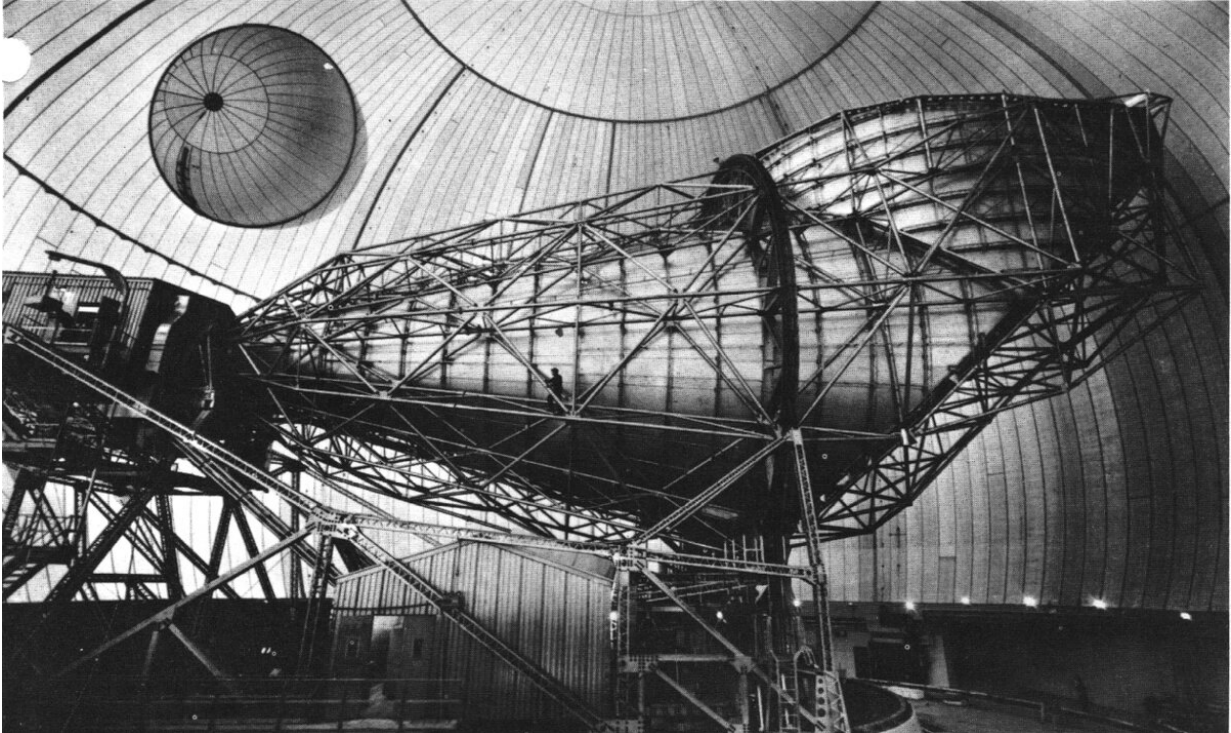


Figure C2.8: Relay 1 Ground Station Antenna — Early Satellite Communications Link Budget

Chapter 3: Semiconductors



Figure C3.1: Single-Crystal Silicon Boule and Wafer



Figure C3.2: Compound-Semiconductor Laser Diode Die (Shown Against a Penny for Scale)

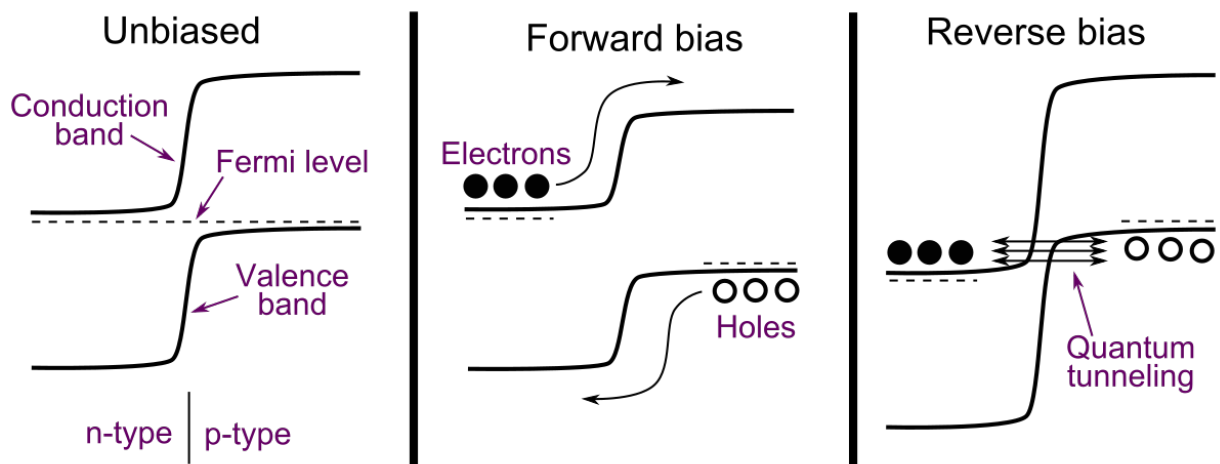


Figure C3.3: Energy-Band Diagram of a PN Junction — Unbiased, Forward-Bias, and Reverse-Bias Conditions

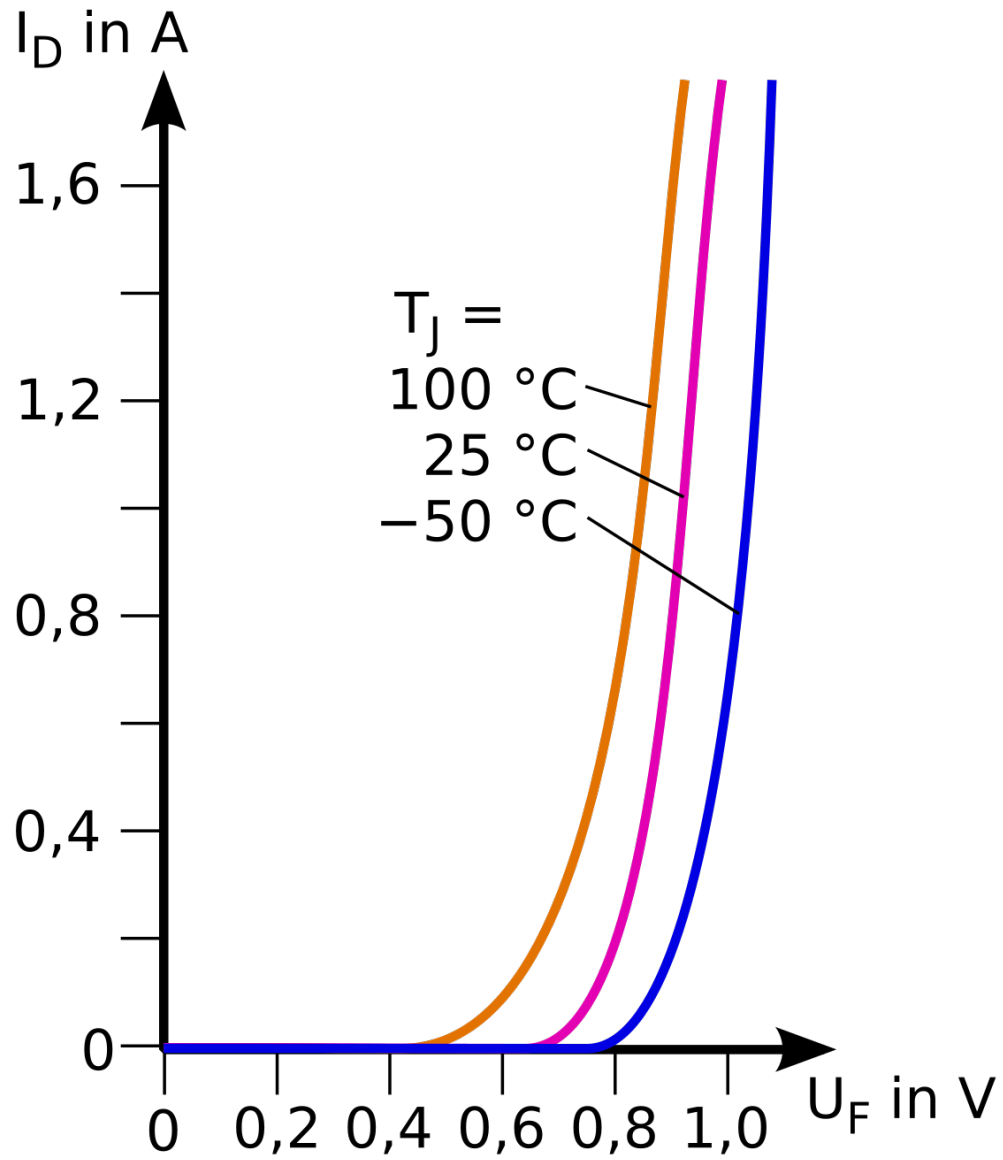


Figure C3.4: Measured I-V Characteristic Curve of a 1N4001 Diode



Figure C3.5: Replica of the First Point-Contact Transistor (Bell Labs, 1947)

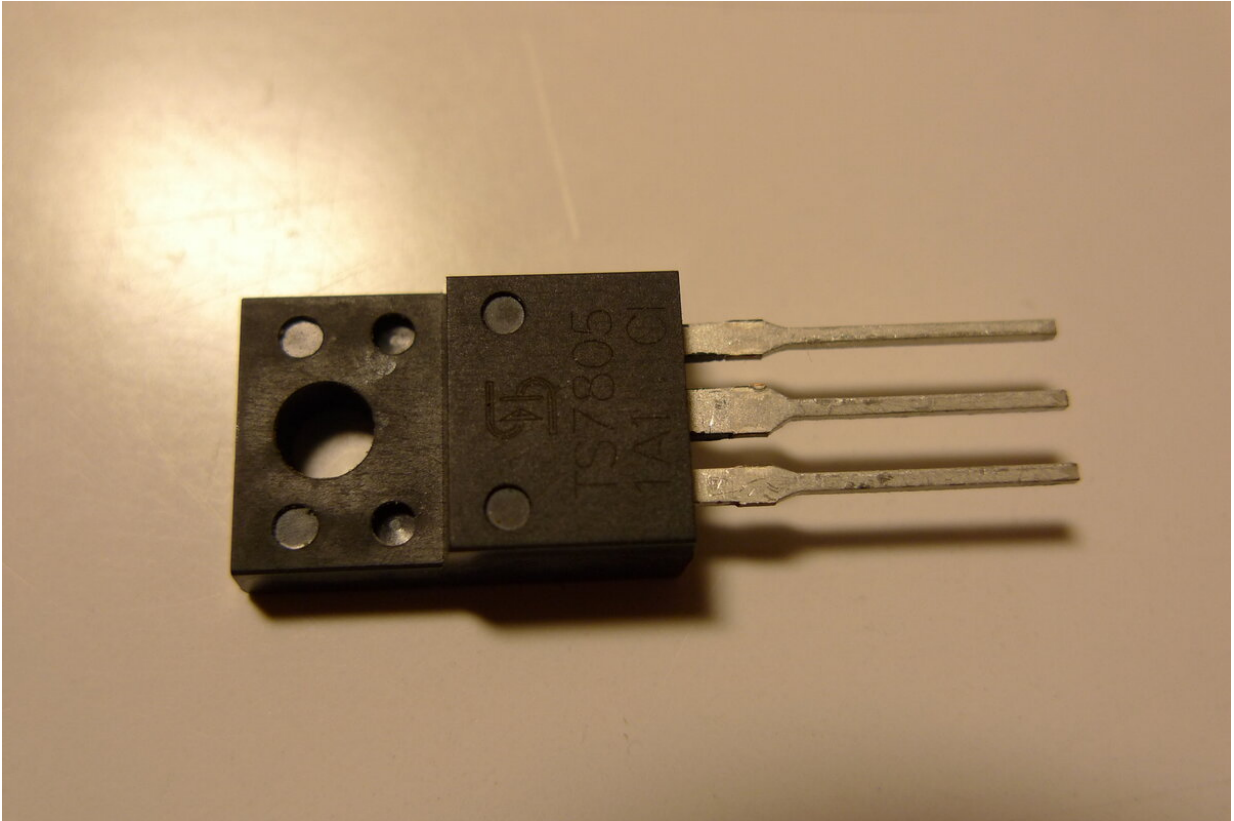


Figure C3.6: TS7805 Linear Voltage Regulator in a TO-220 Package



Figure C3.7: Etched Silicon Wafer Showing Individual Die

Chapter 4: Control Systems

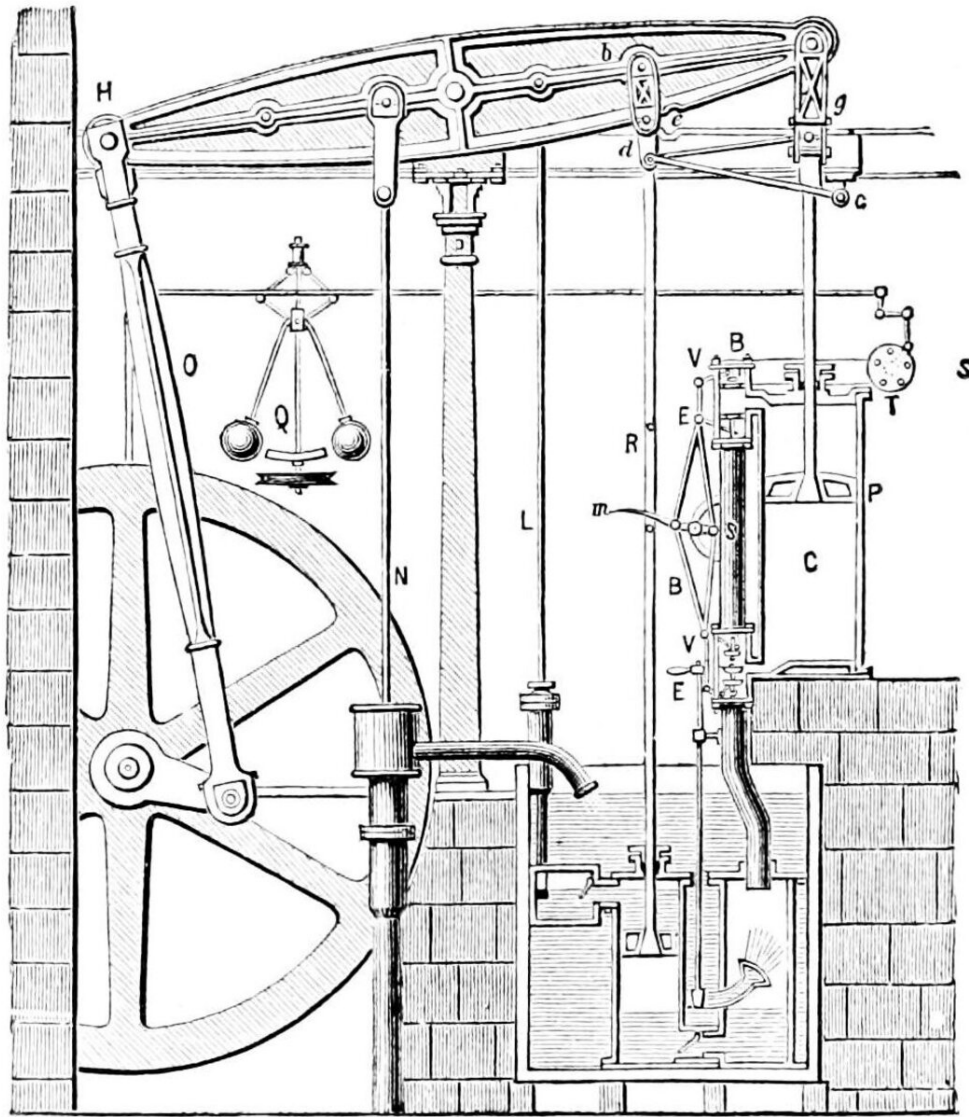


Figure C4.1: Watt Steam Engine (1780) — Uncontrolled (Open-Loop) Mechanism

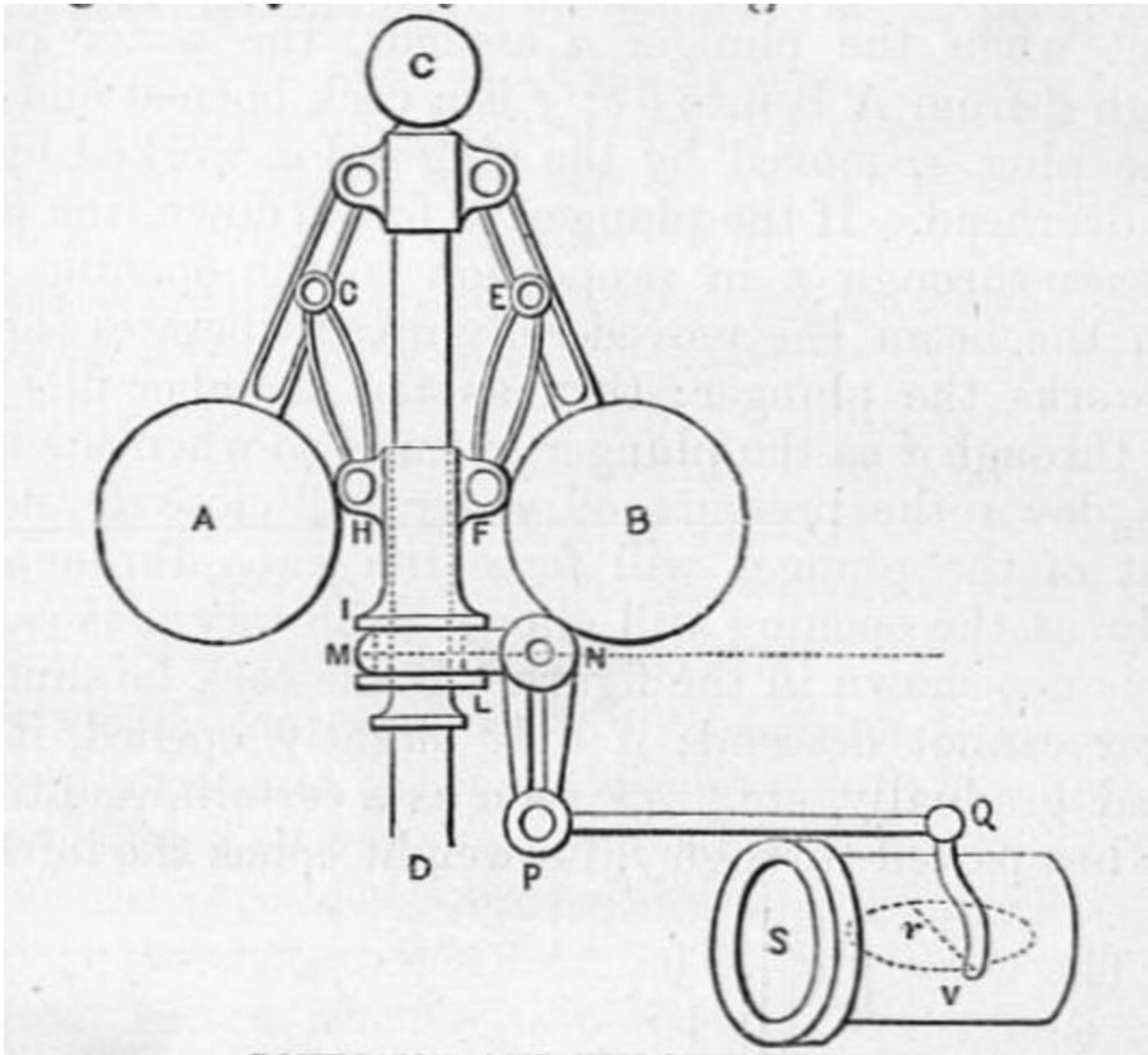


Figure C4.2: Watt's Centrifugal Flyball Governor — the First Widely Used Closed-Loop Feedback Controller

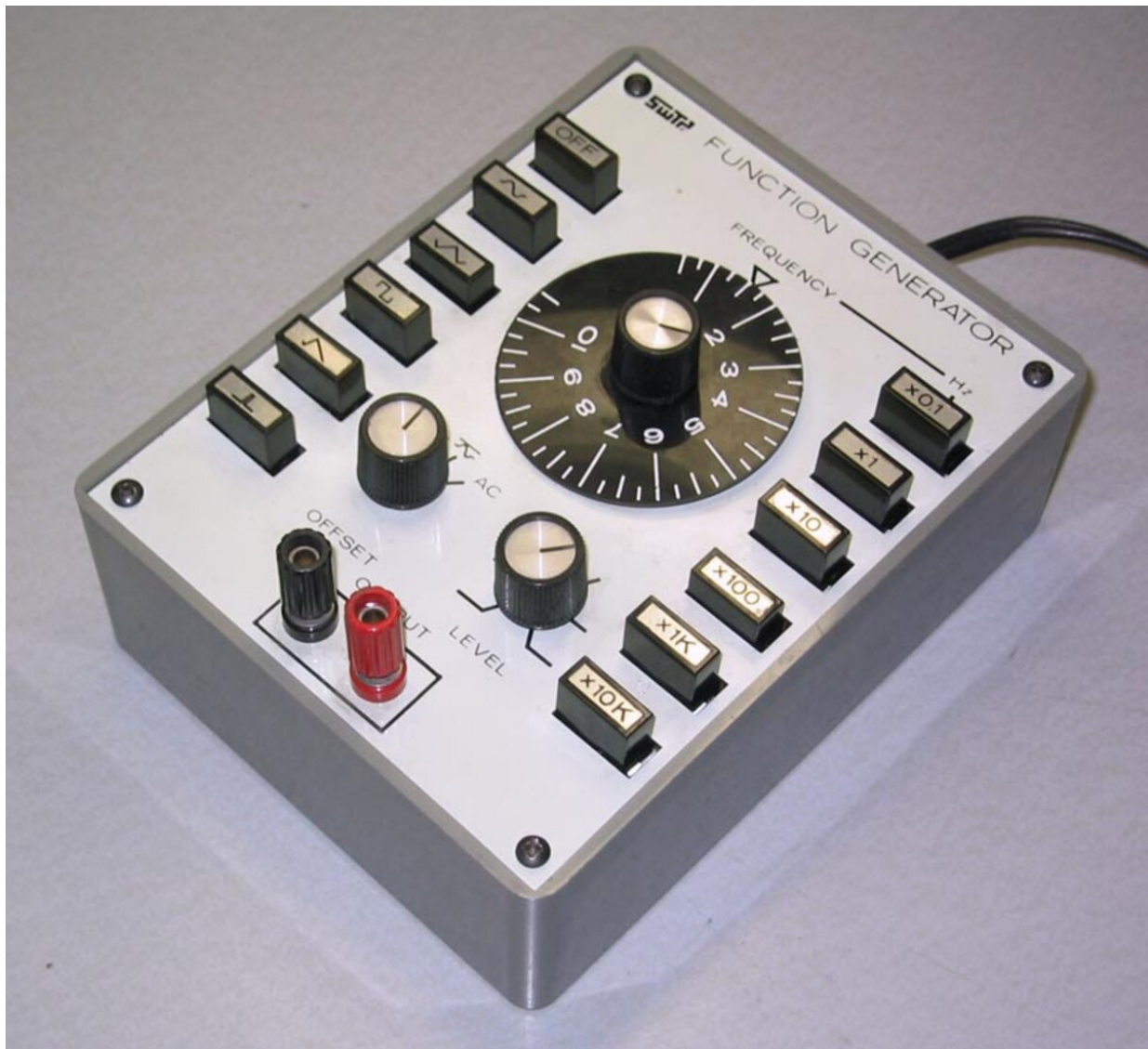


Figure C4.3: Function Generator for Producing Reference Control Signals

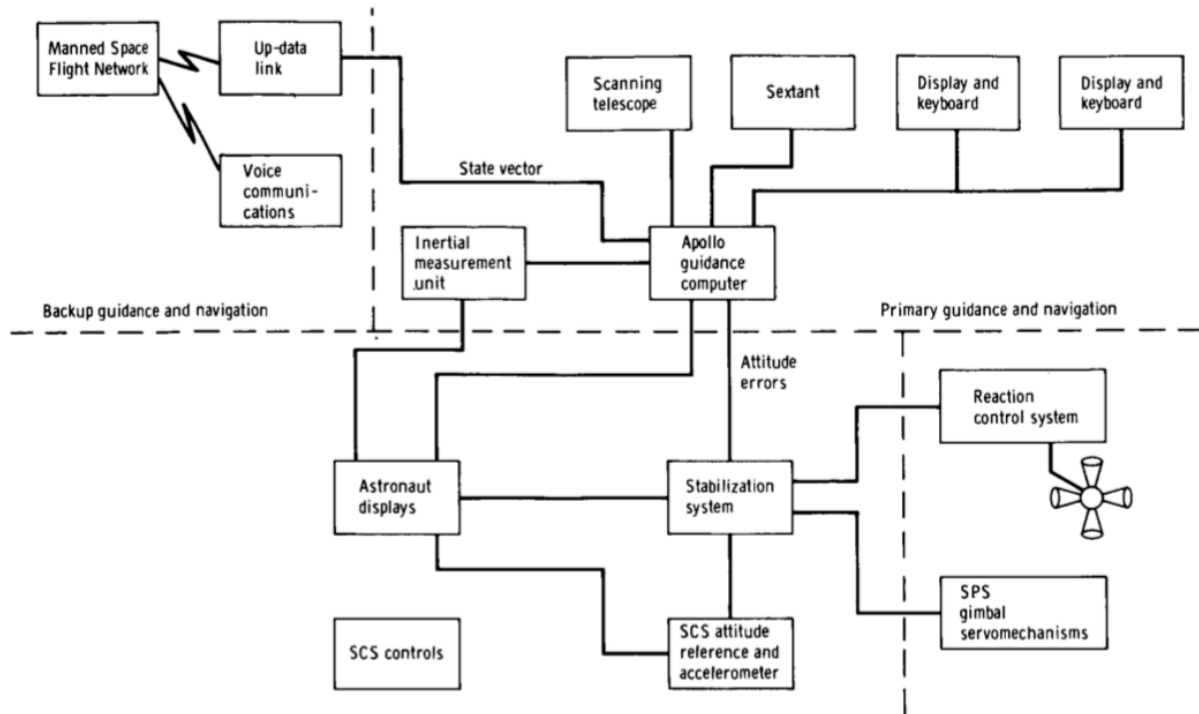


Figure C4.4: Apollo Block I Guidance System Block Diagram

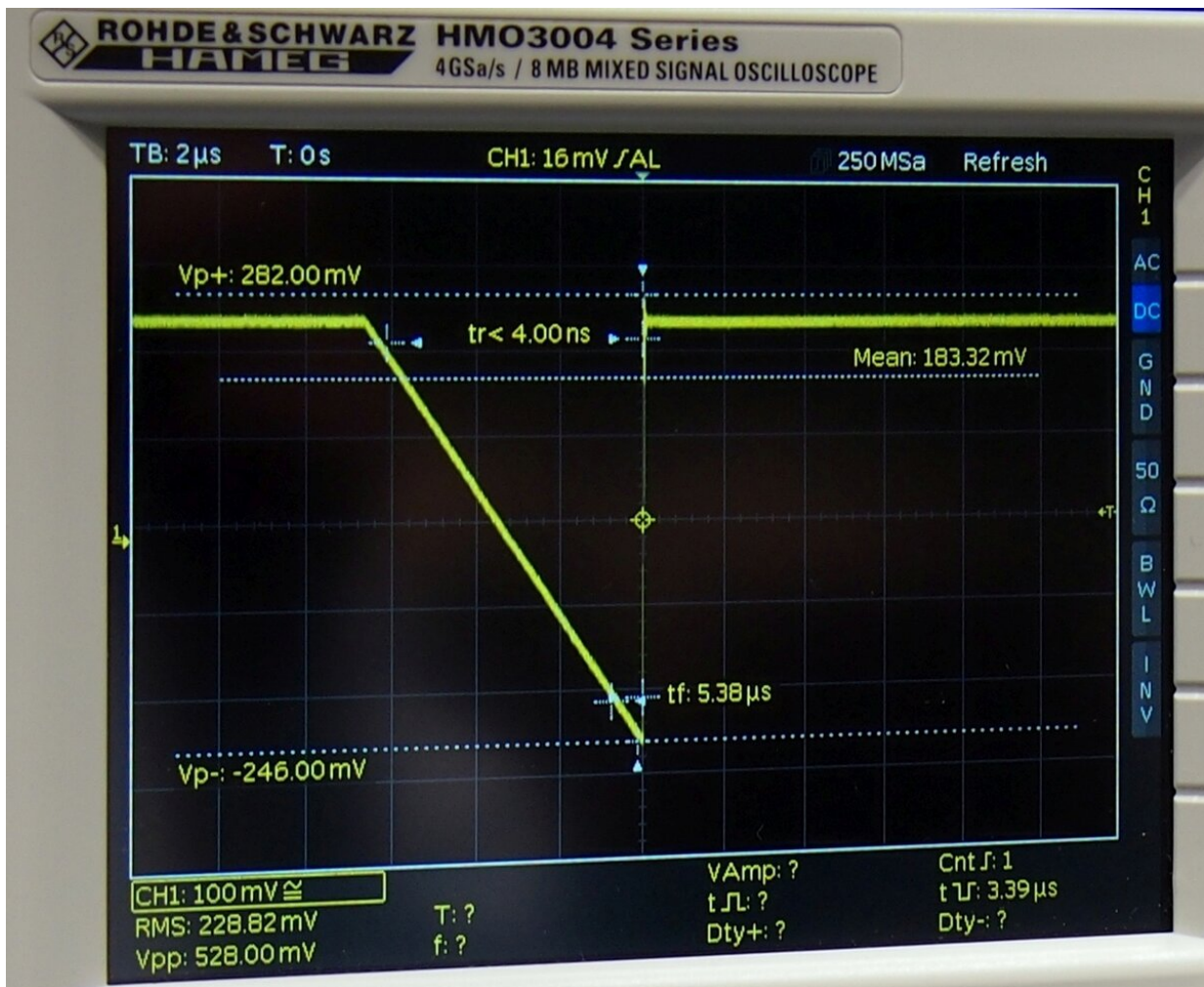


Figure C4.5: Oscilloscope Capture of a Time-Domain Step/Fall Response



Figure C4.6: Honeywell Round Thermostat (1953) — Classic Bang-Bang/PID Feedback Control

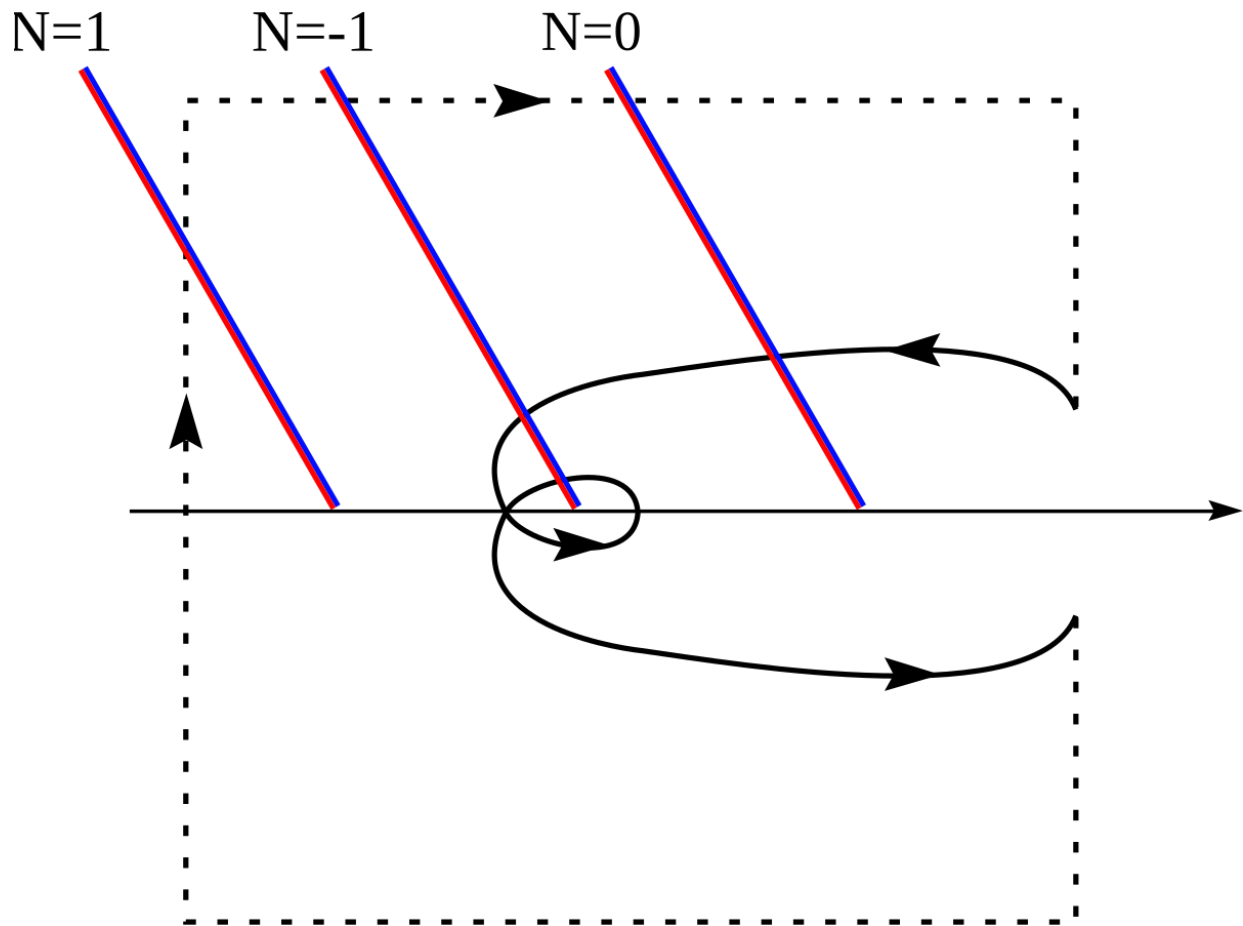


Figure C4.7: Nyquist Stability Criterion Diagram

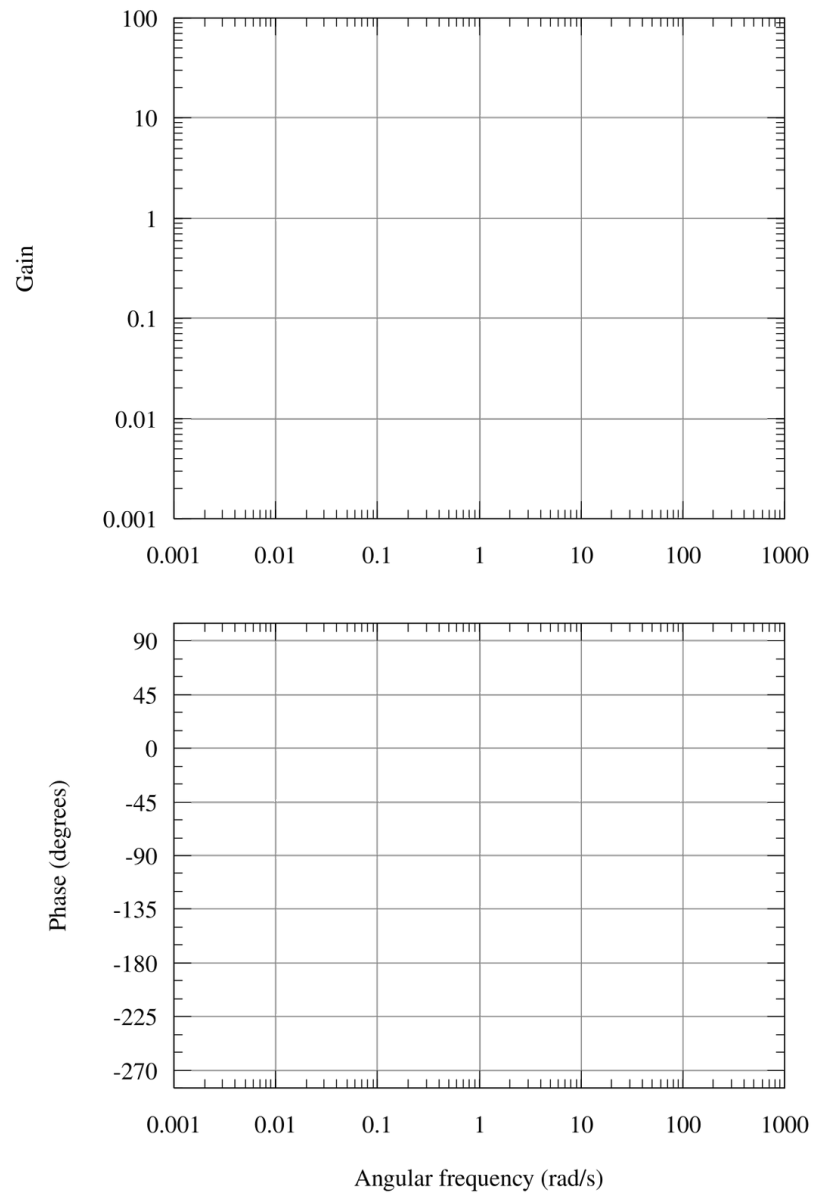


Figure C4.8: Bode Plot Template — Magnitude and Phase vs. Frequency

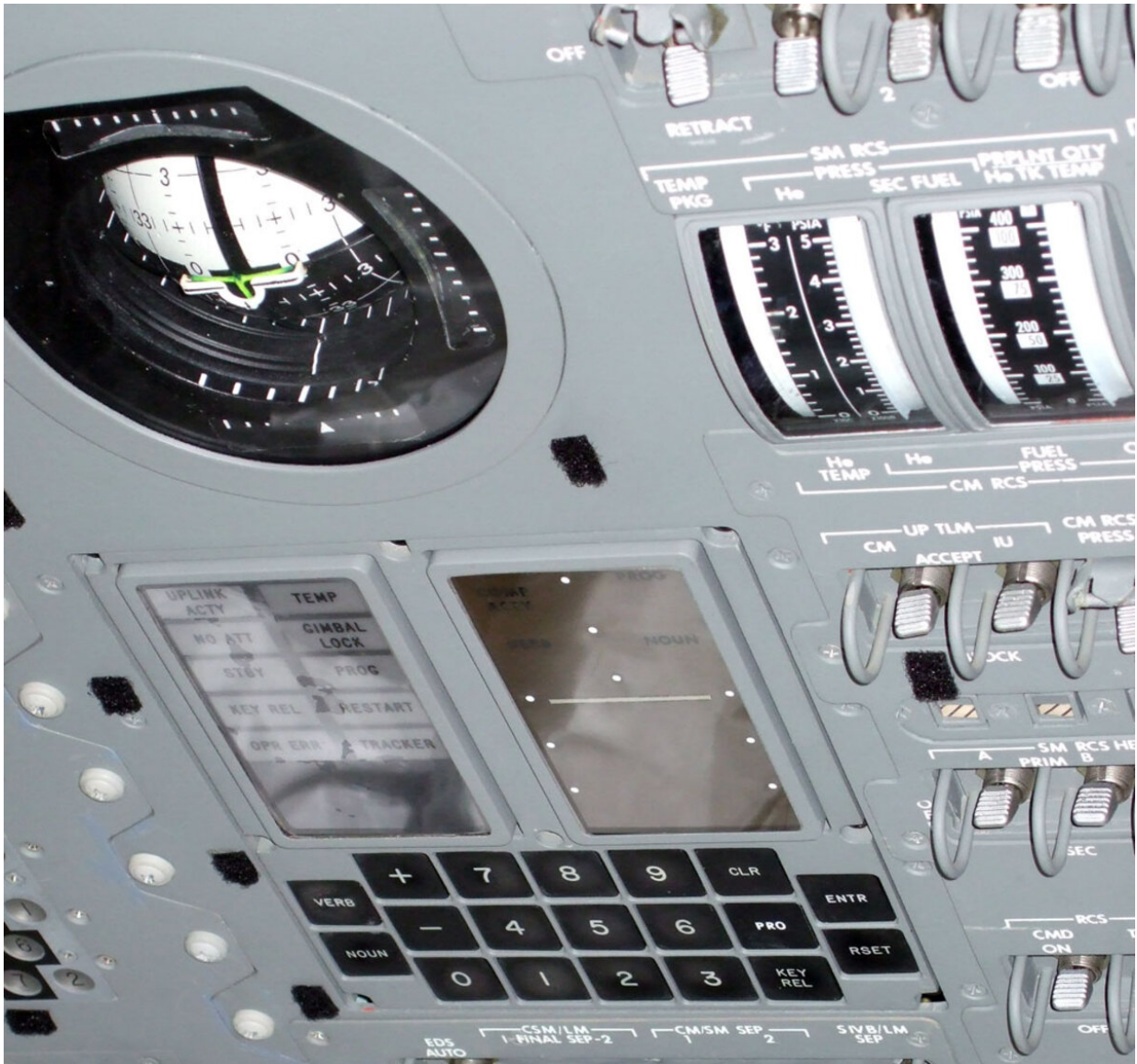


Figure C4.9: Apollo Guidance Computer DSKY (Display and Keyboard) — Onboard State Estimation and Control Interface

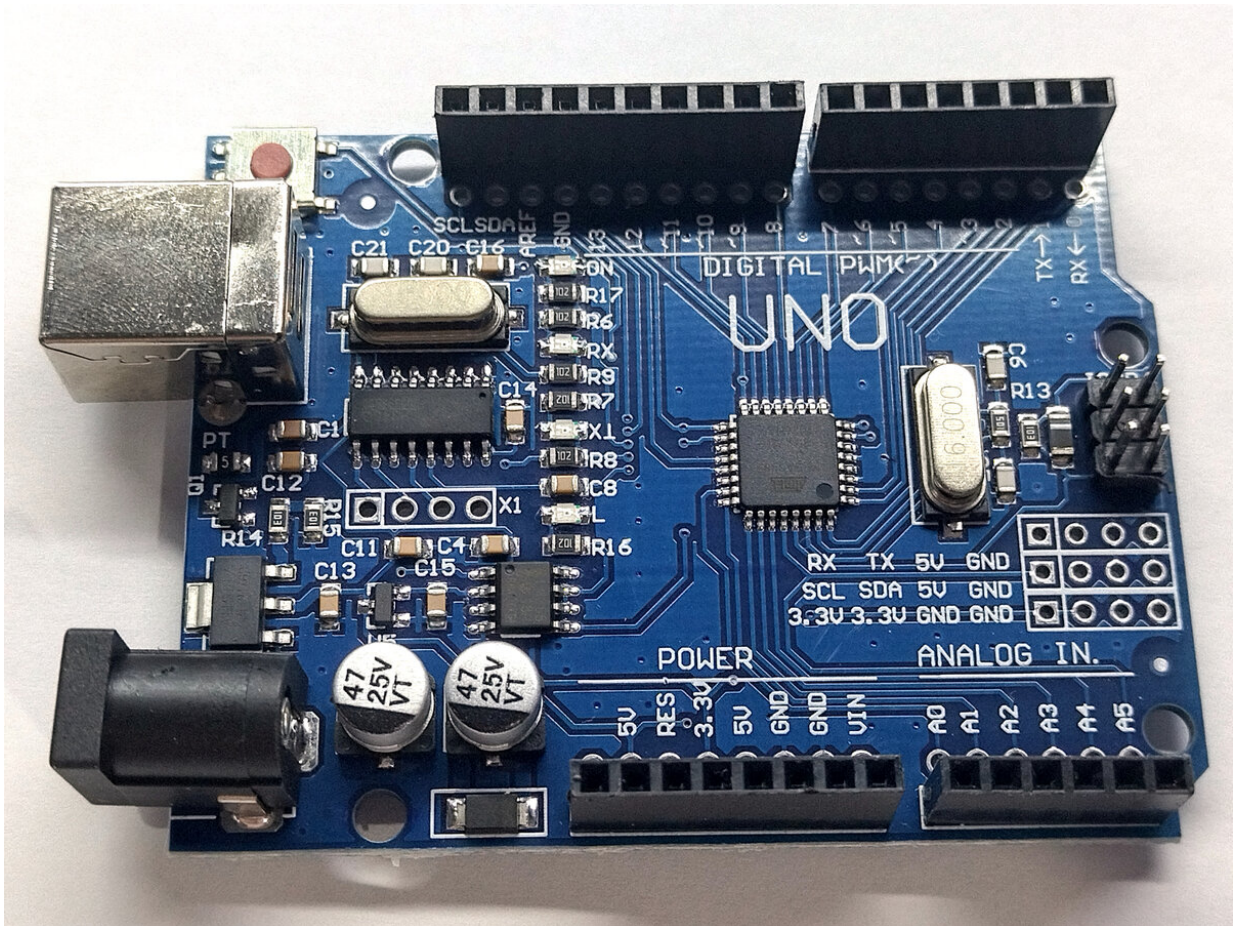


Figure C4.10: Arduino Uno Microcontroller Board — a Common Digital Control System Platform

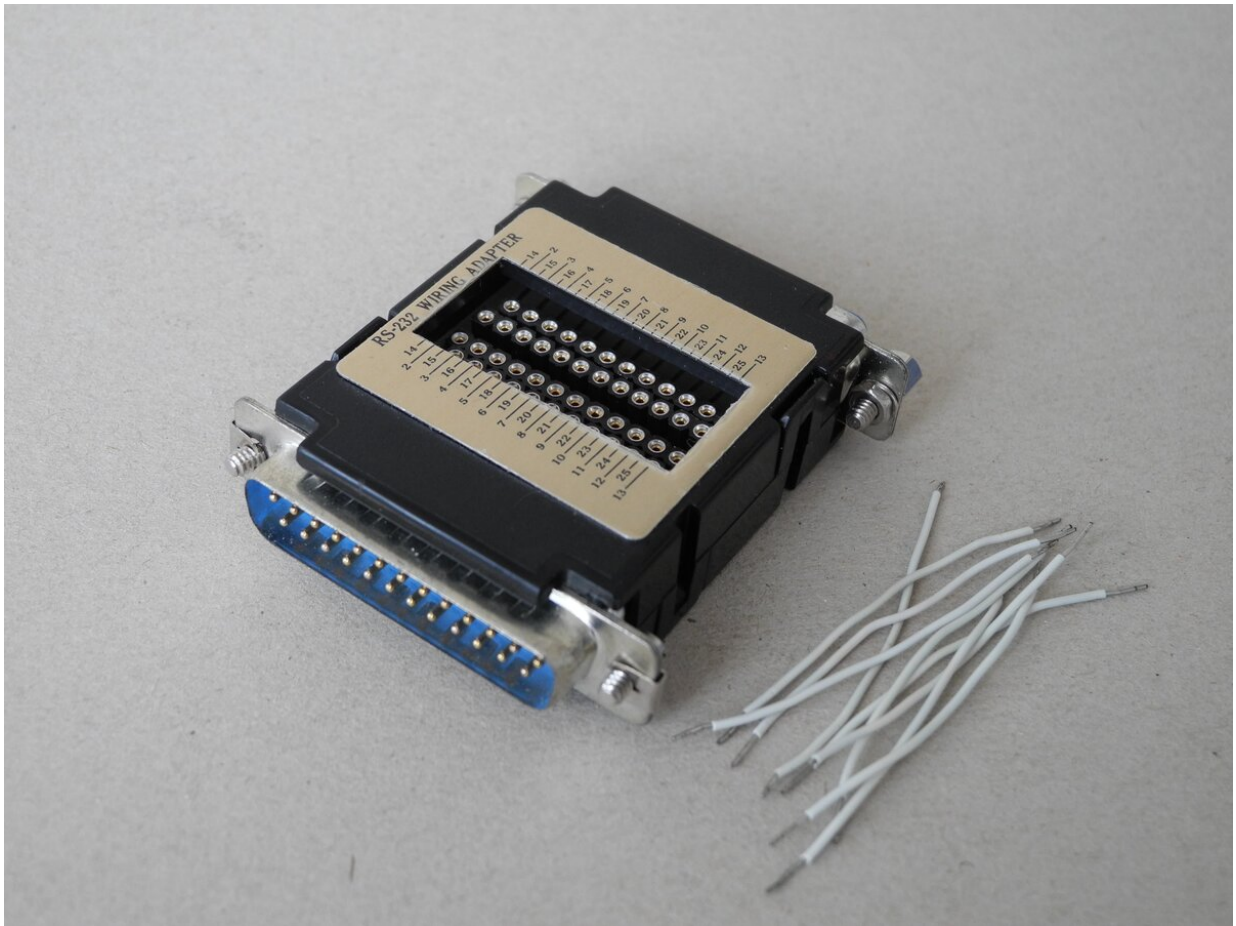


Figure C5.2: RS-232 Wiring Adapter (DB-25) for Serial Communication Interfaces

Chapter 6: Digital Logic



Figure C6.1: ERA 1103 Computer — Early Binary/Digital Calculating Machine (1955)

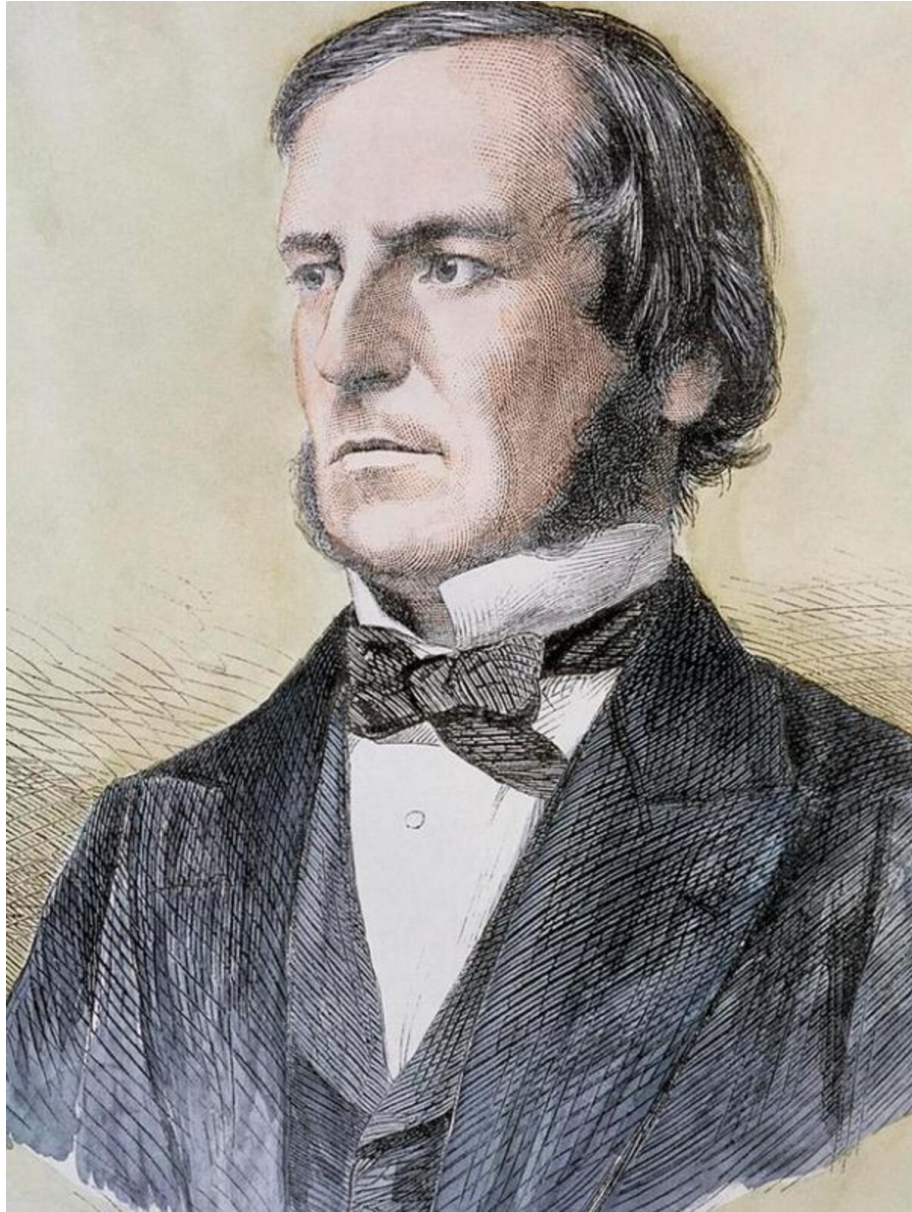


Figure C6.2: George Boole, Originator of Boolean Algebra

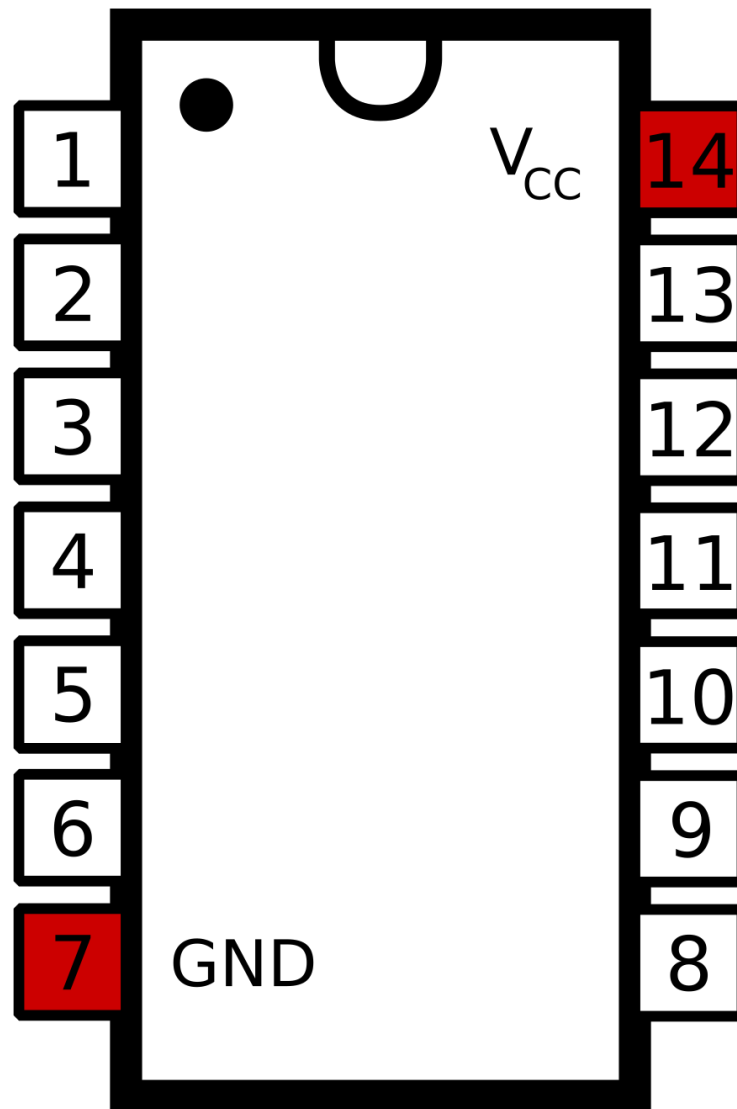


Figure C6.3: Generic 14-Pin TTL Logic Gate IC Pinout

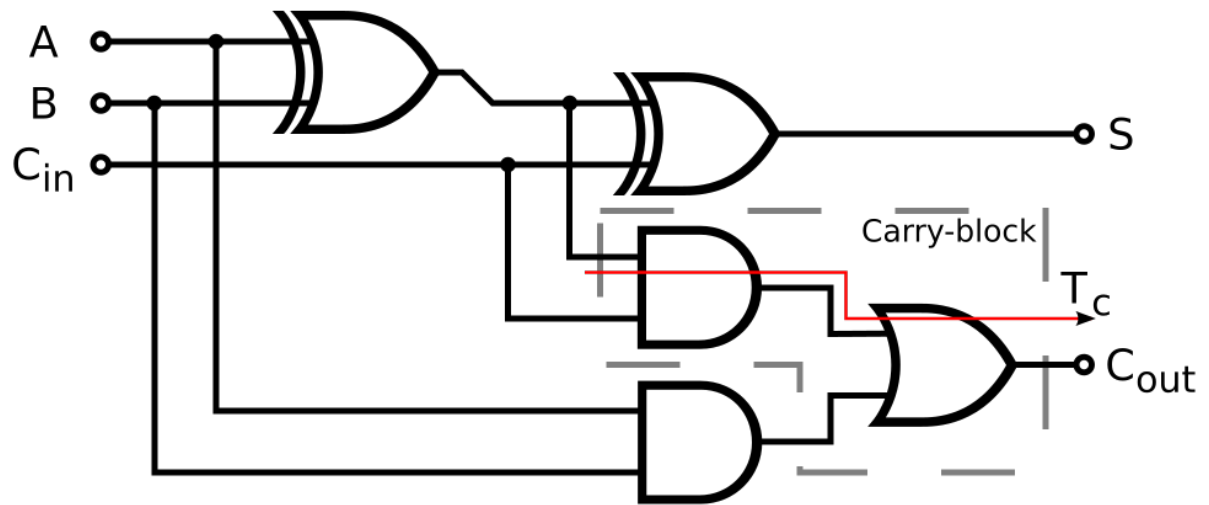


Figure C6.4: Full-Adder Combinational Logic Diagram

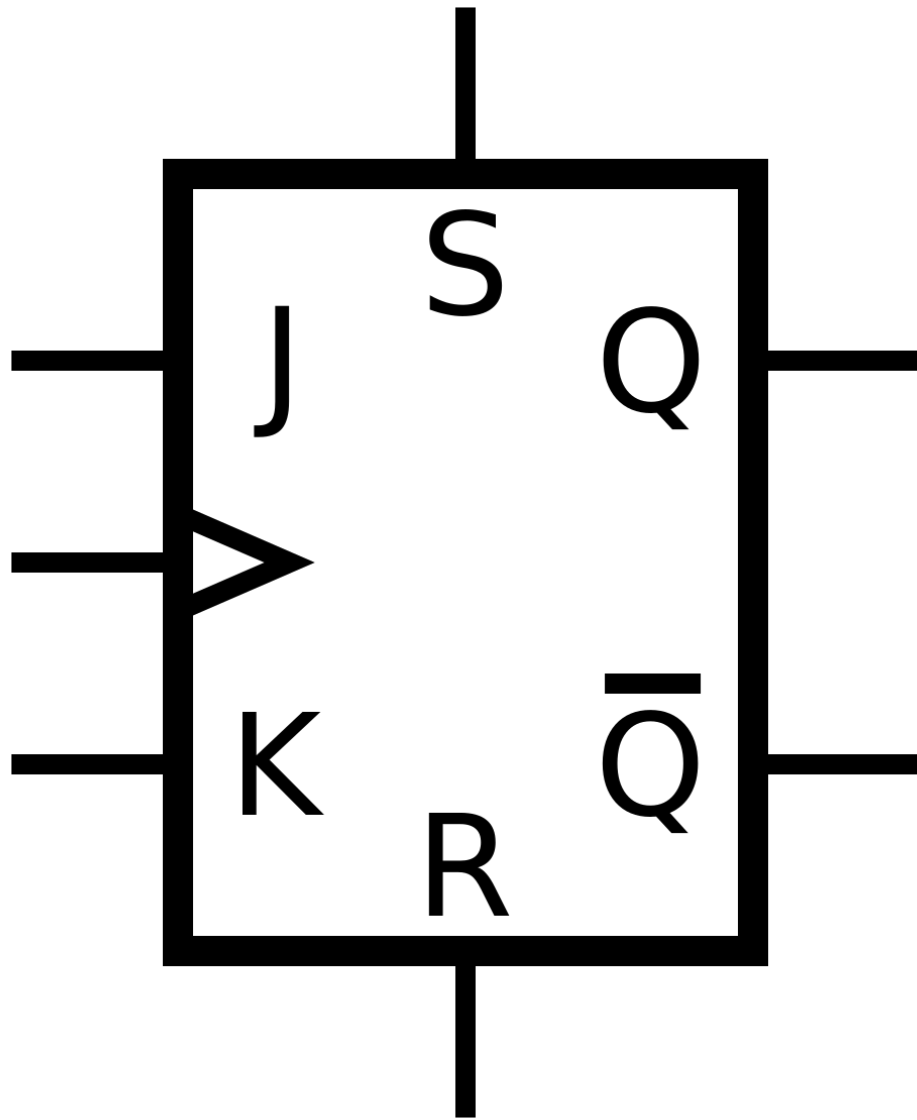


Figure C6.5: JK Flip-Flop Sequential Logic Diagram

Chapter 7: Circuit Analysis



Figure C7.1: Benjamin Franklin Drawing Electricity from the Sky (Benjamin West, c. 1816)



Figure C7.3: Georg Simon Ohm, Namesake of Ohm's Law

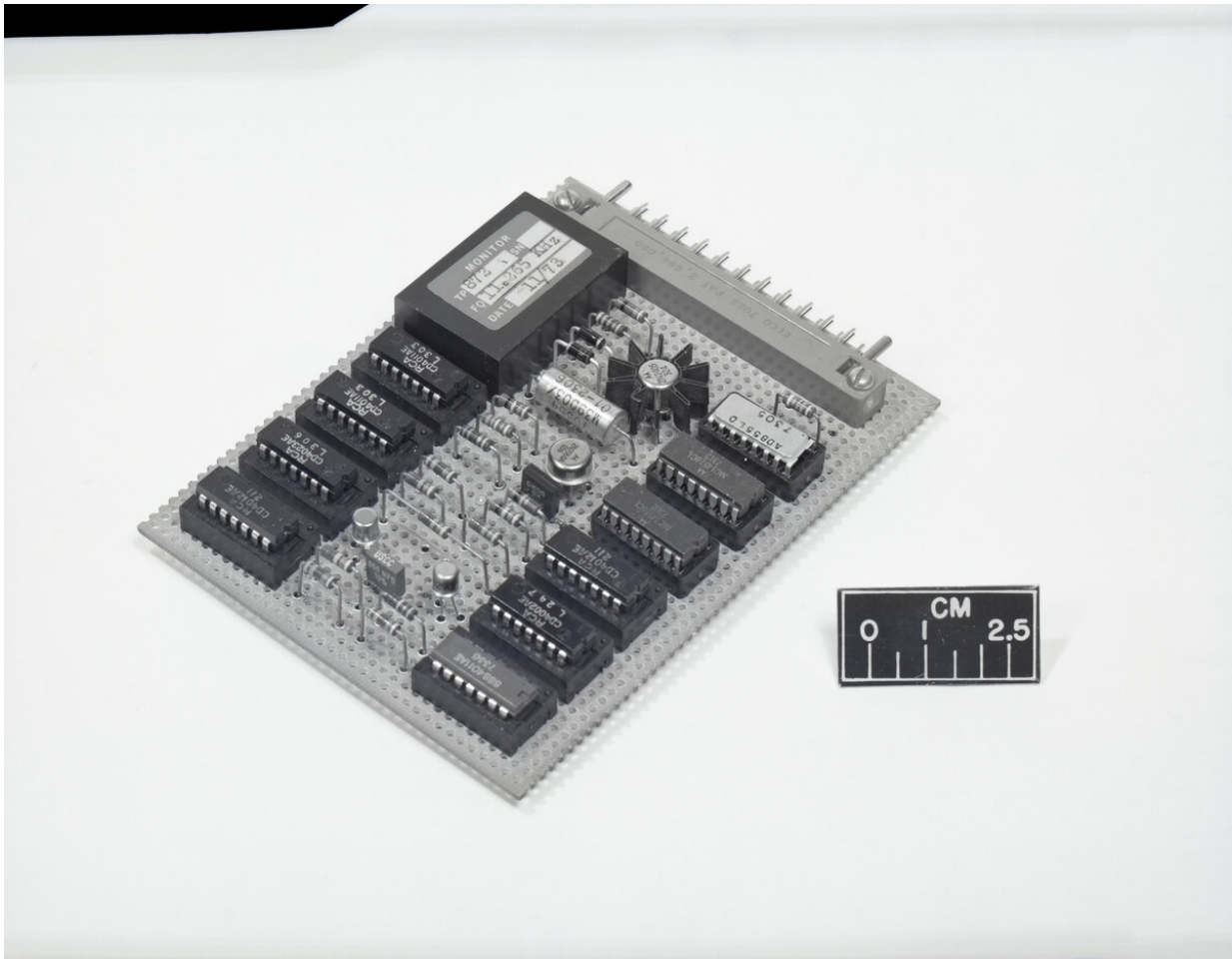


Figure C7.4: Populated DC Circuit Board on Breadboard-Style Perfboard

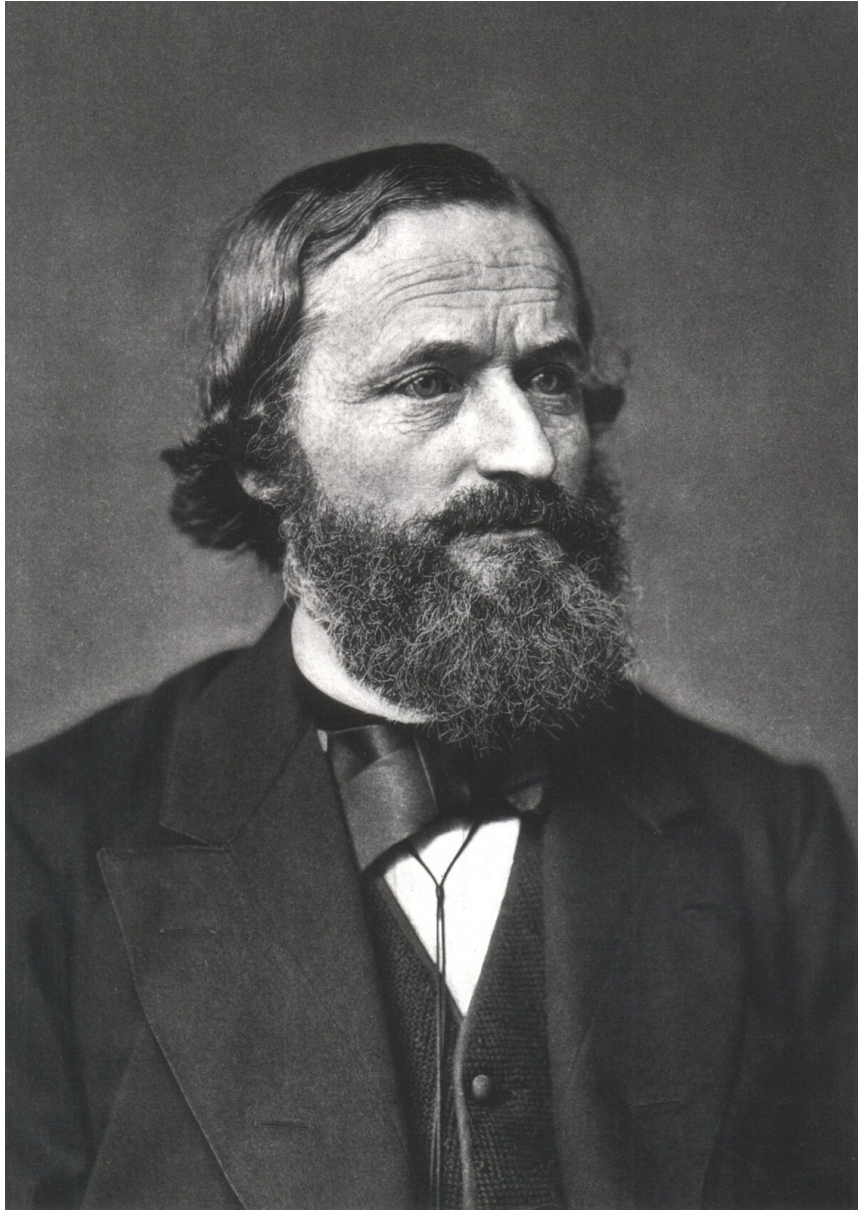


Figure C7.5: Gustav Robert Kirchhoff, Originator of Kirchhoff's Circuit Laws



Figure C7.6: Léon Charles Thévenin, Namesake of Thévenin's Theorem

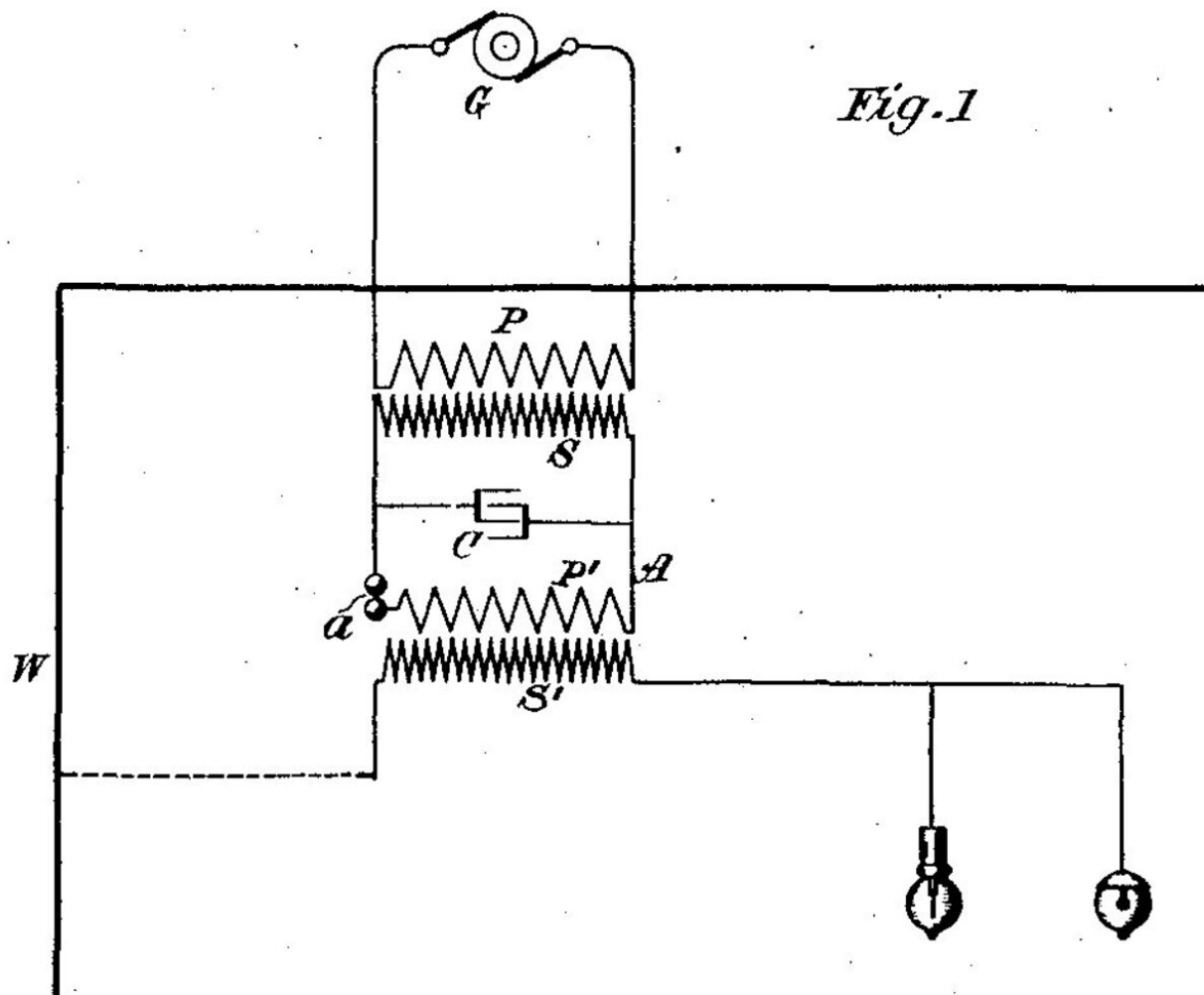


Figure C7.7: Nikola Tesla's 1891 Patent Drawing for a System of Electric (AC) Lighting

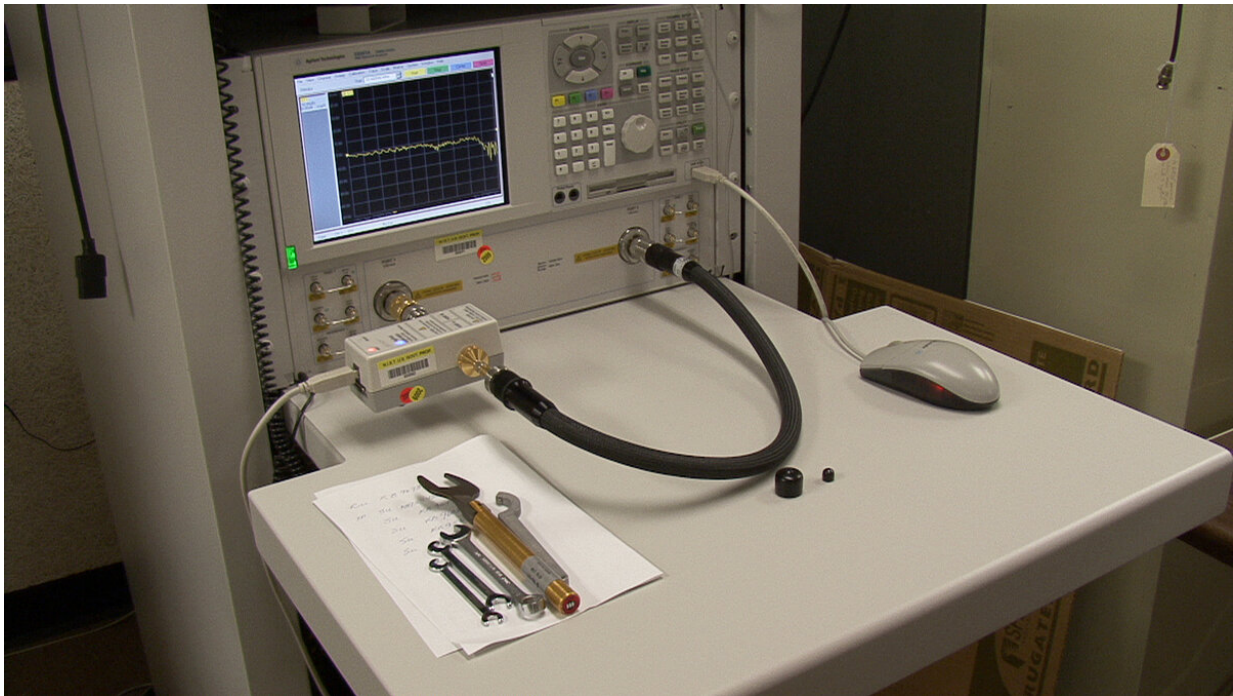


Figure C7.9: Vector Network Analyzer Used for Two-Port Network Calibration

Chapter 8: Signal Processing

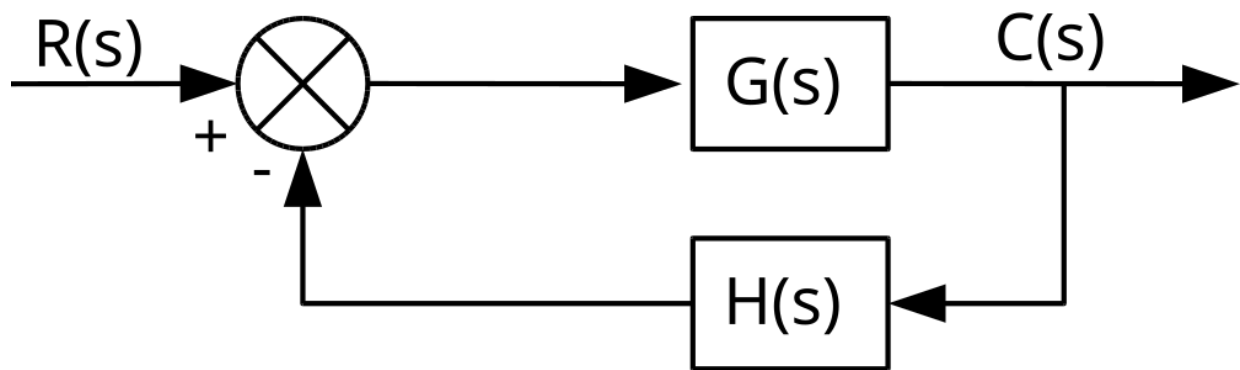


Figure C8.1: General Closed-Loop Feedback System Block Diagram



Figure C8.2: Joseph Fourier, Originator of Fourier Analysis



Figure C8.3: Pierre-Simon Laplace, Namesake of the Laplace Transform

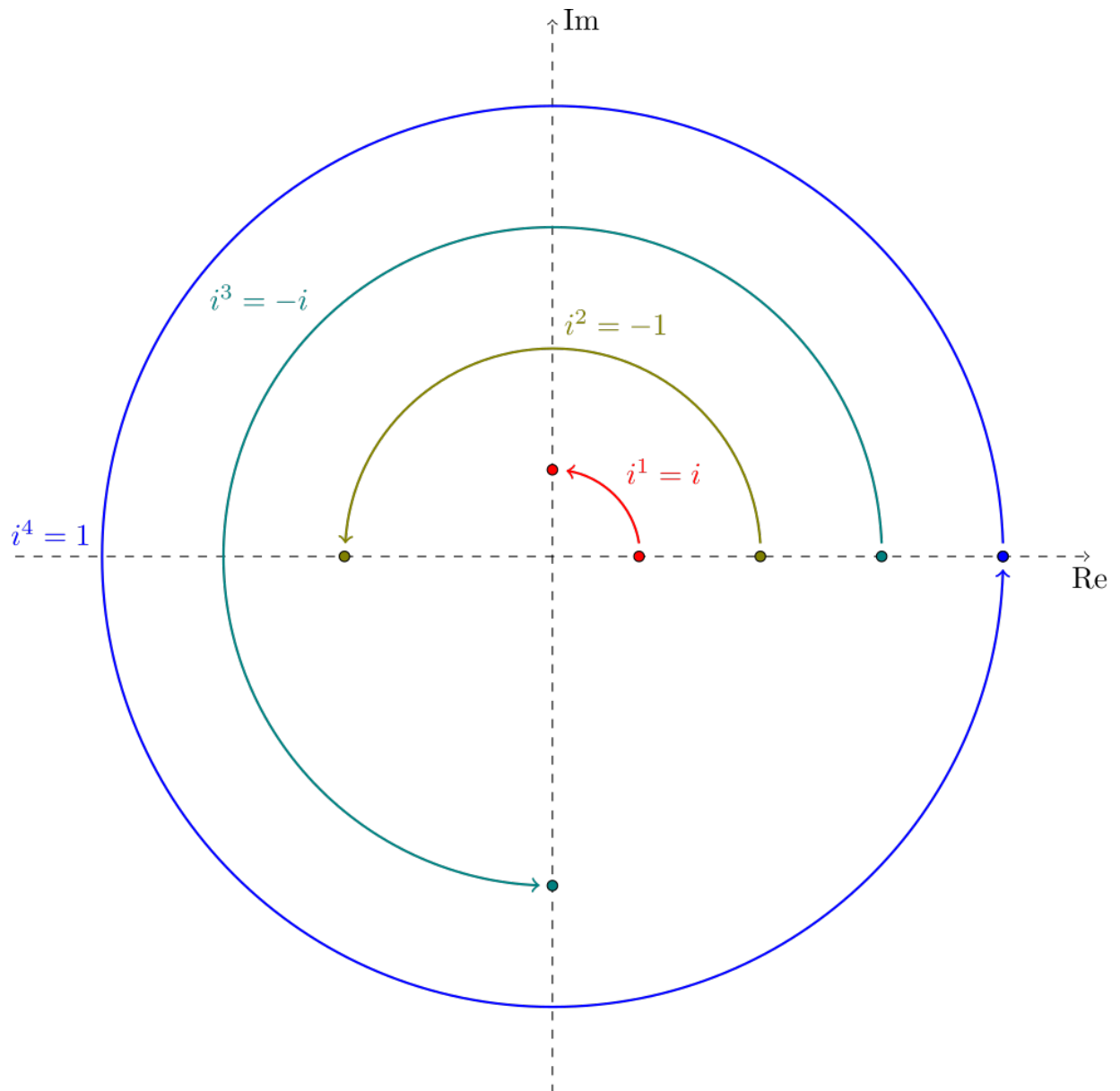


Figure C8.4: Rotations on the Complex Plane — Basis for the Z-Plane Used in the Z-Transform

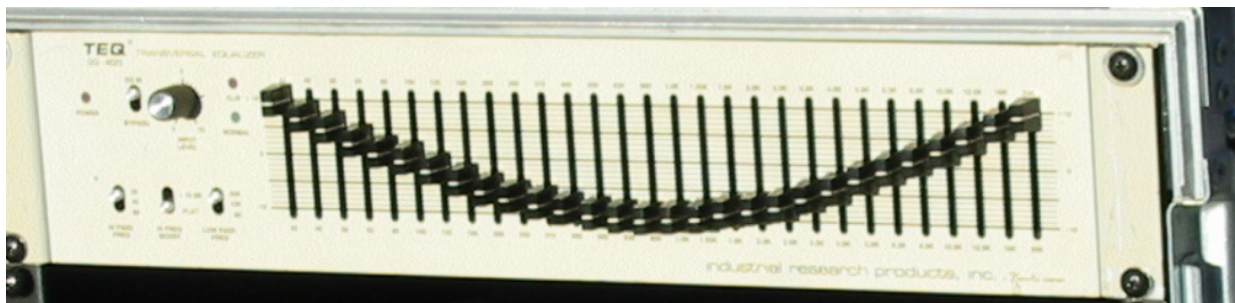


Figure C8.5: Graphic Equalizer — a Bank of Adjustable Digital/Analog Filters

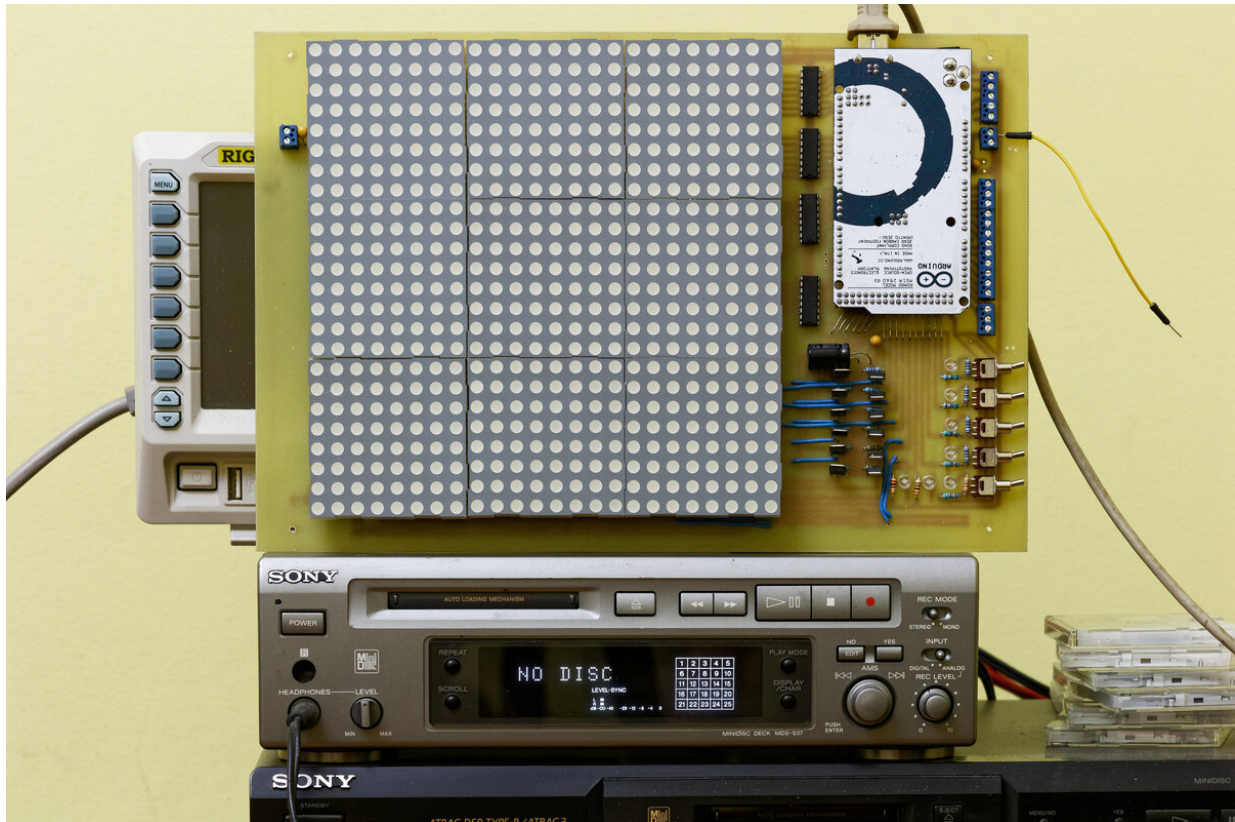


Figure C8.6: DIY Arduino-Based LED Audio Spectrum Analyzer

Chapter 9: Electromagnetics



Figure C9.1: Van de Graaff Generator — Electrostatic Charge Demonstration

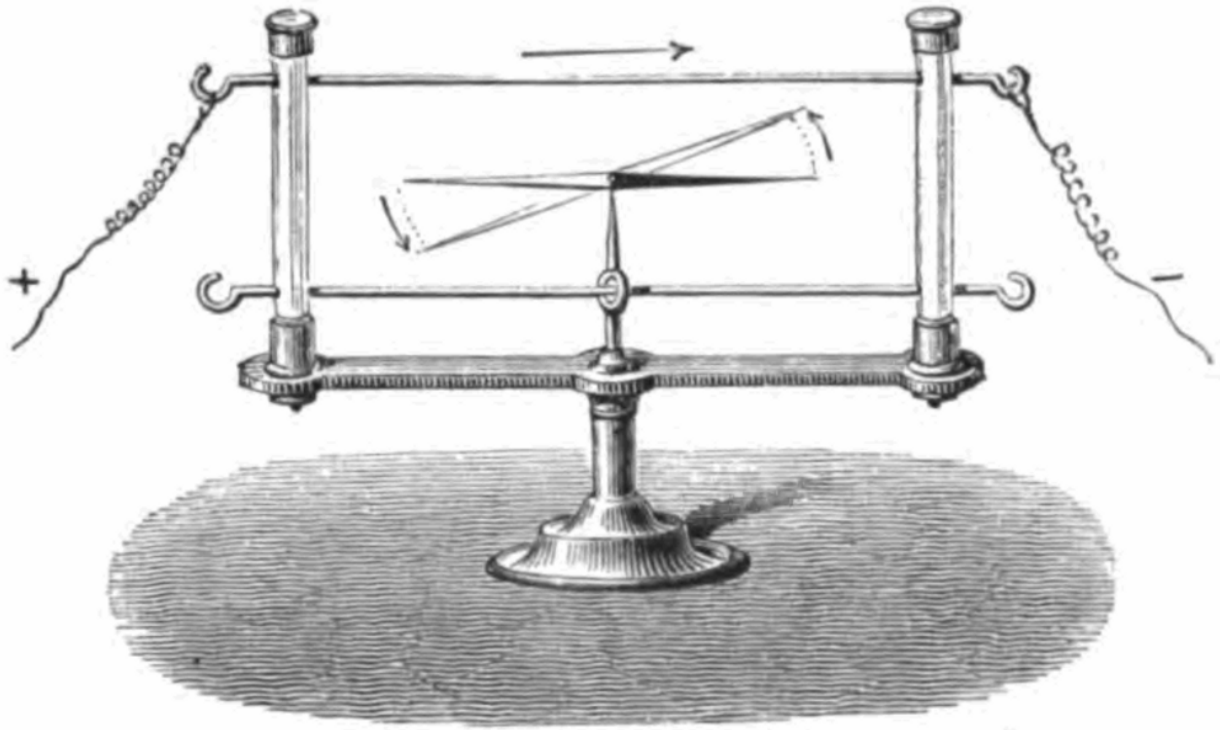


Figure C9.2: Ørsted's Experiment — Current-Carrying Wire Deflecting a Compass Needle



Figure C9.3: James Clerk Maxwell, Formulator of Maxwell's Equations



Figure C9.4: Heinrich Hertz, First to Demonstrate Electromagnetic Waves

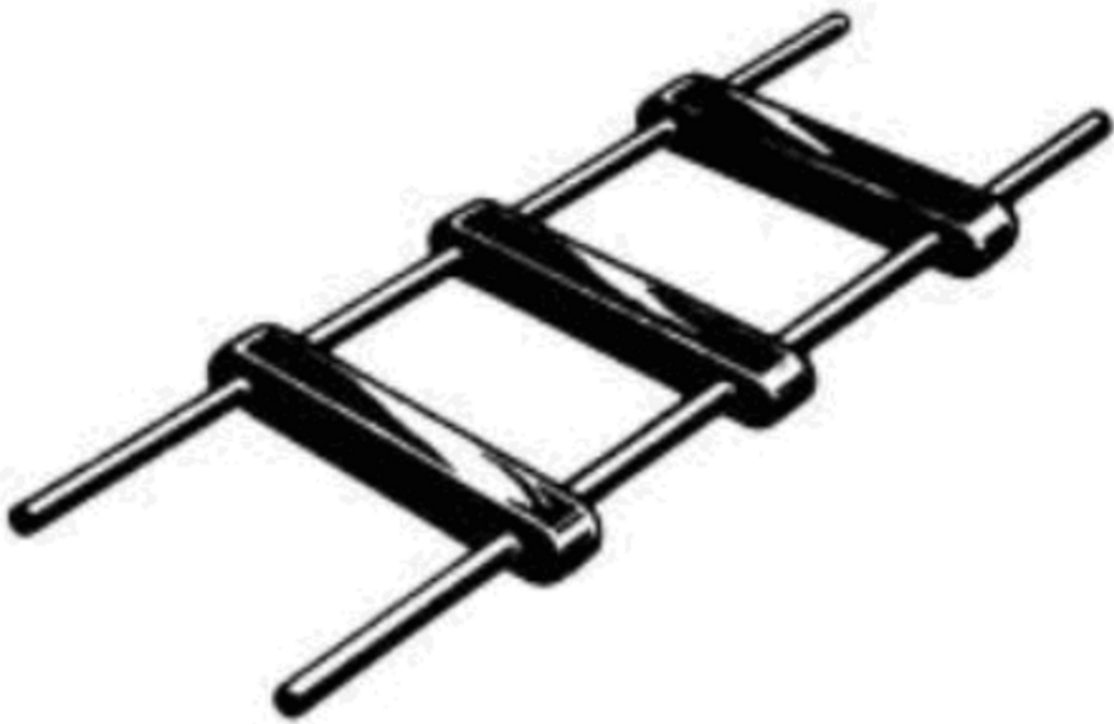


Figure C9.5: Ladder Line — an Open-Wire Balanced Transmission Line

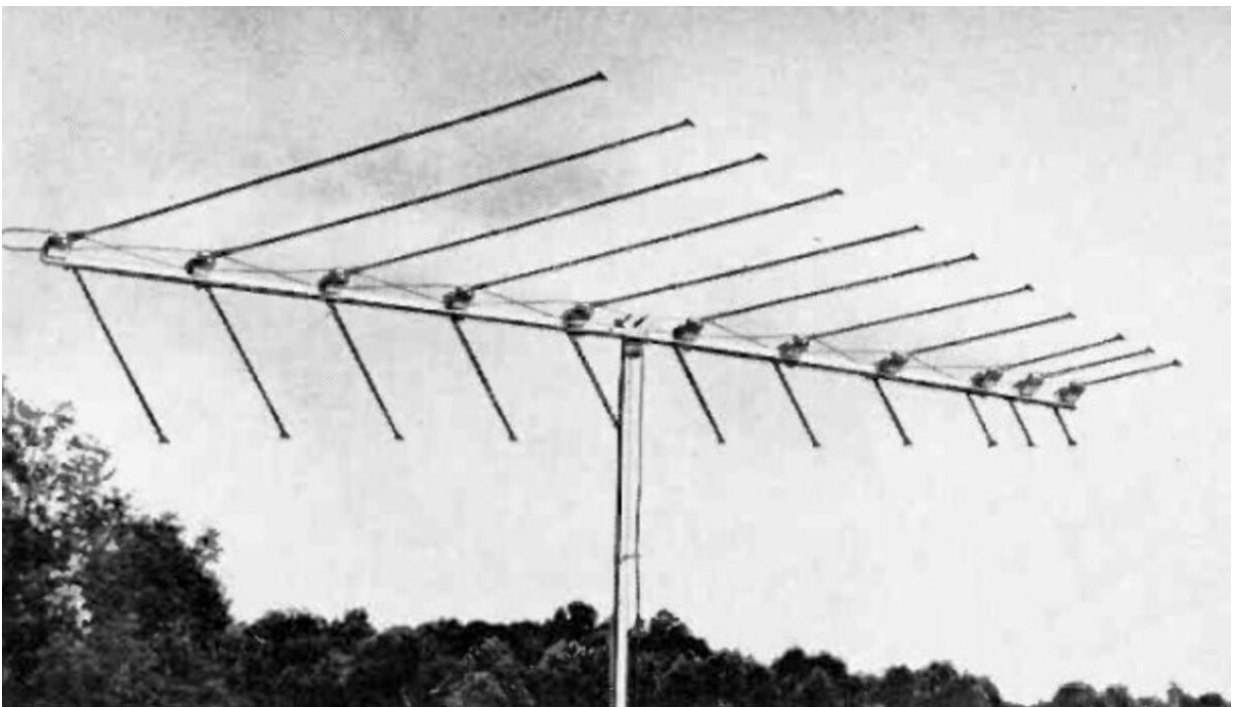


Figure C9.6: Log-Periodic VHF TV Antenna (1963)

Chapter 10: Power Electronics

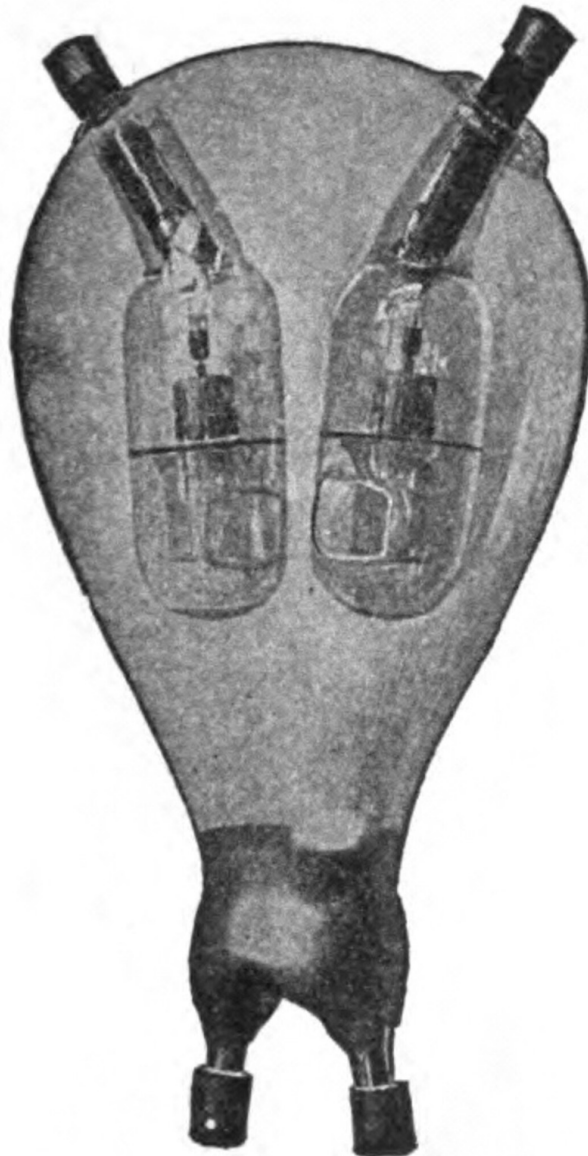


Figure C10.2: Twin-Anode Mercury-Arc Rectifier Bulb

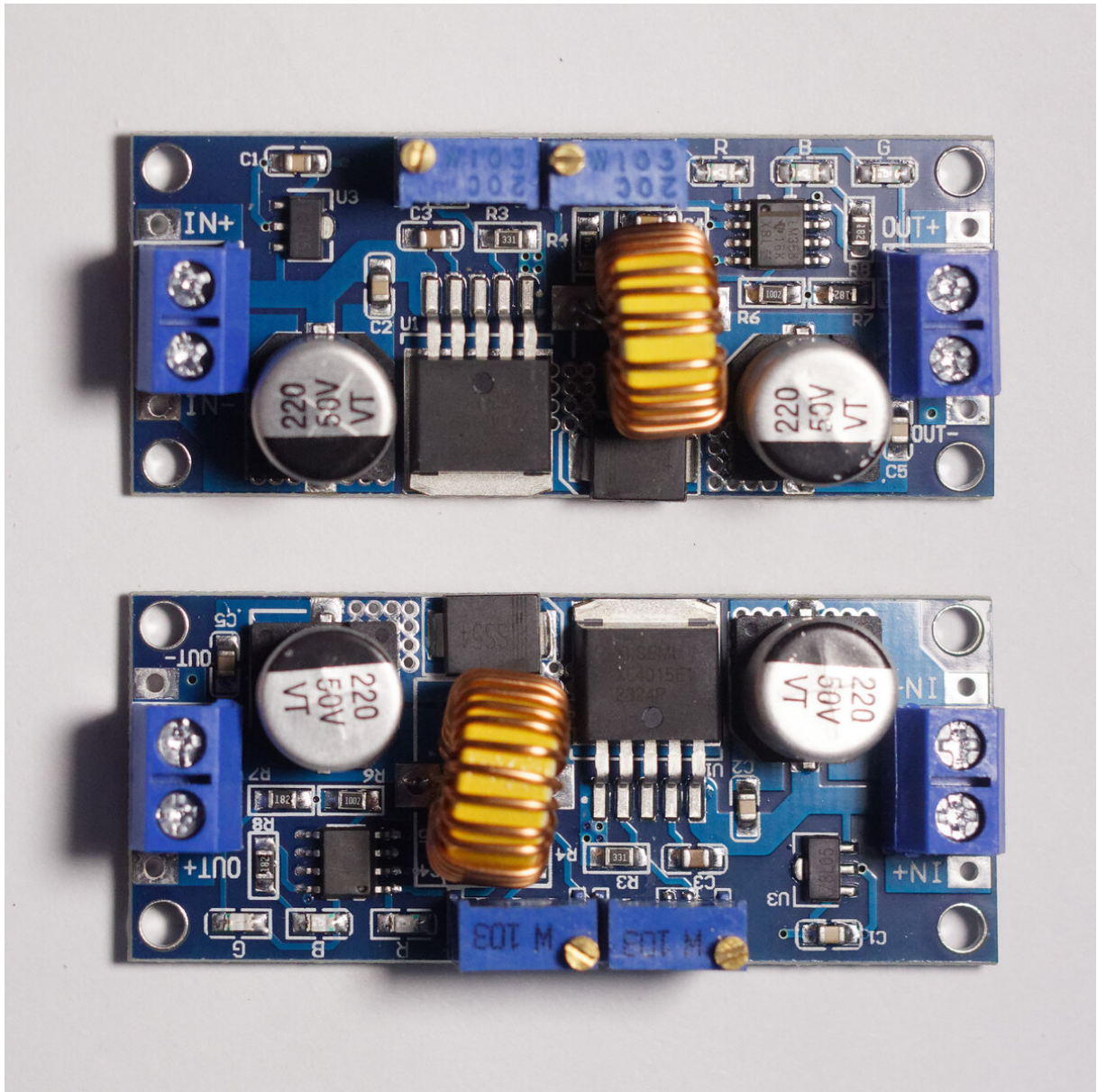


Figure C10.3: XL4015-Based CC/CV Buck Converter Modules

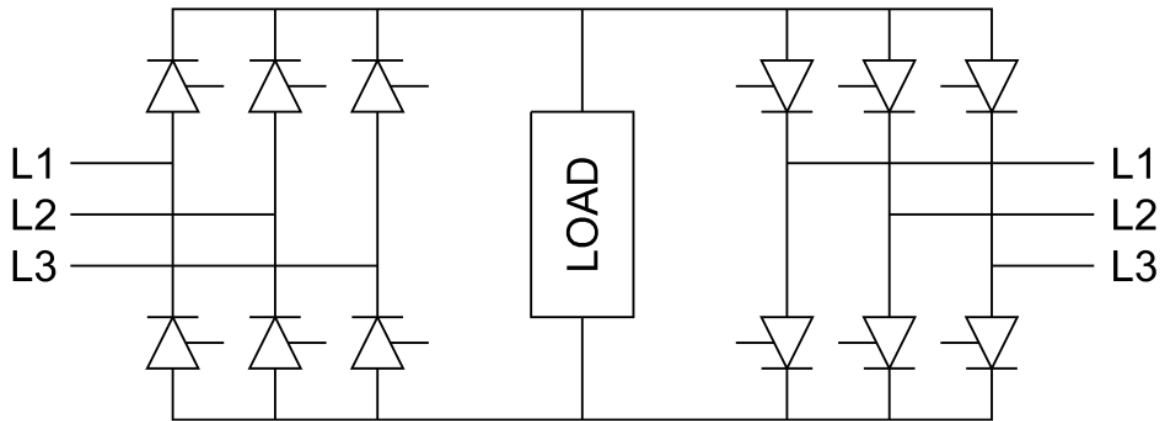


Figure C10.5: Cycloconverter Circuit Diagram (Direct AC-AC Conversion)

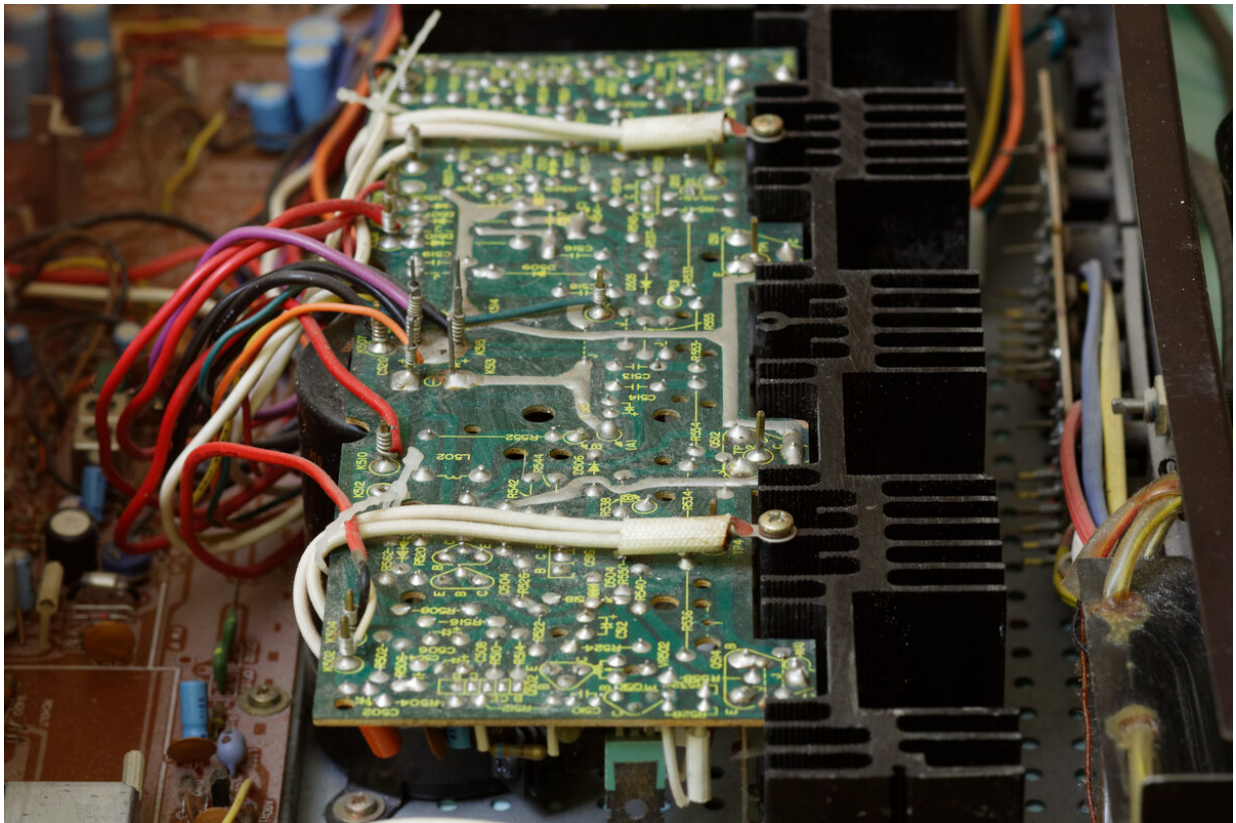


Figure C10.6: Heatsinked Power Amplifier Output Transistors — Thermal Management in Power Electronics



Figure C10.7: Pole-Mounted Power Factor Correction Capacitor Bank

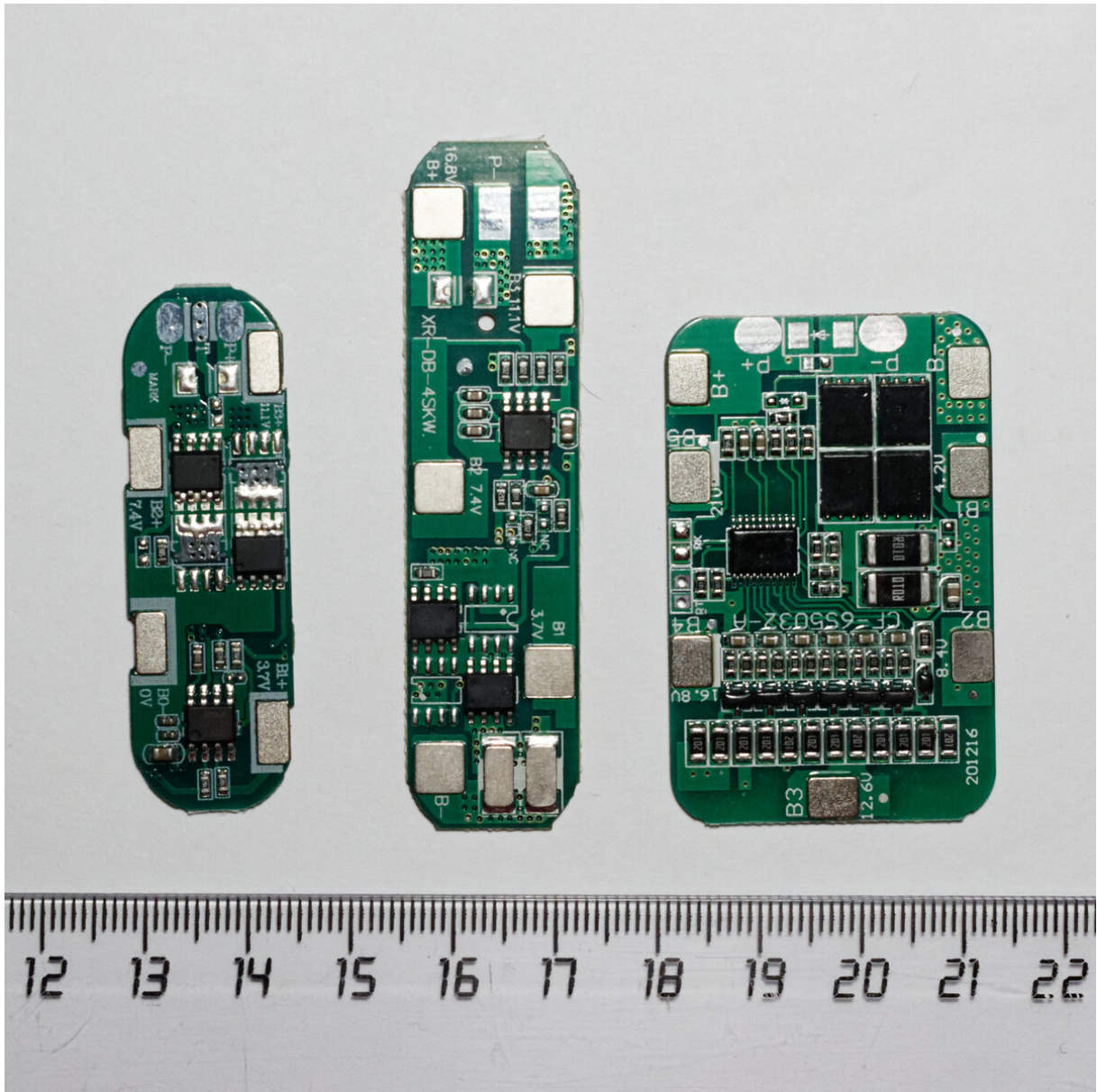


Figure C10.8: Generic Lithium Battery Management System (BMS) Boards, 3S/4S/6S



Figure C10.10: Tesla V3 Supercharger — High-Power DC Fast Charging Station



Figure C10.11: Commercial Supercapacitor (Ultracapacitor) Products

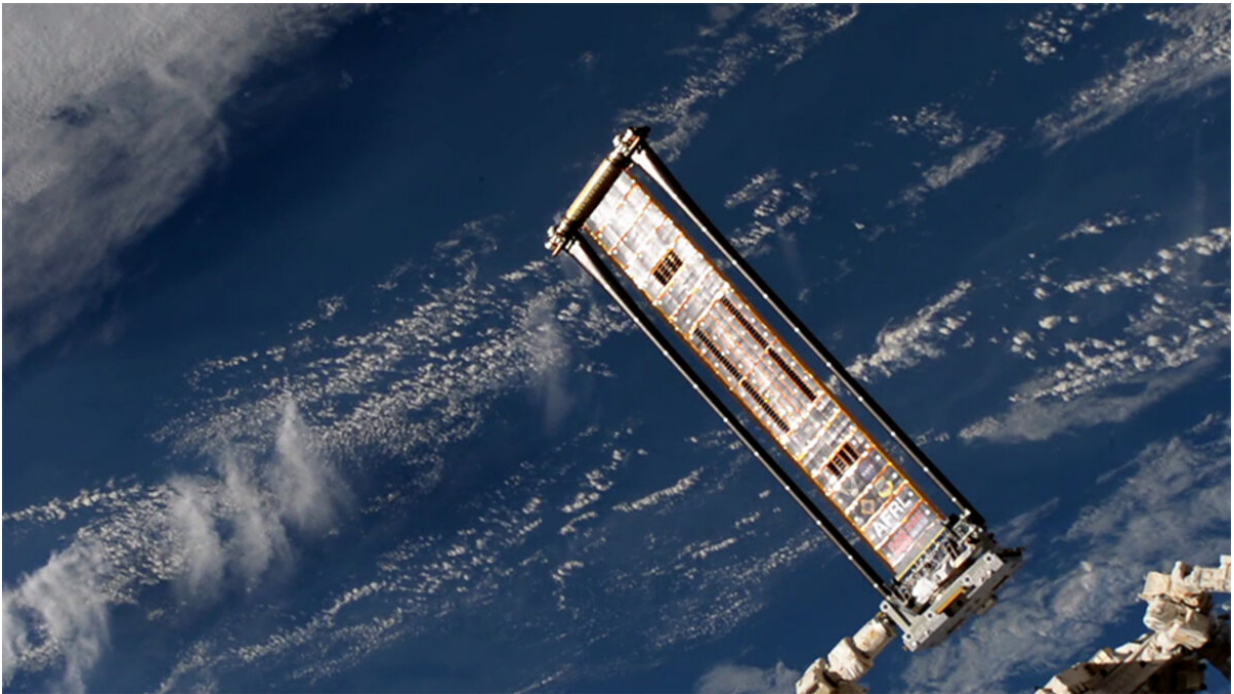


Figure C10.12: Roll-Out Solar Array (ROSA) Deployed on the International Space Station

Chapter 11: Instrumentation and Measurement

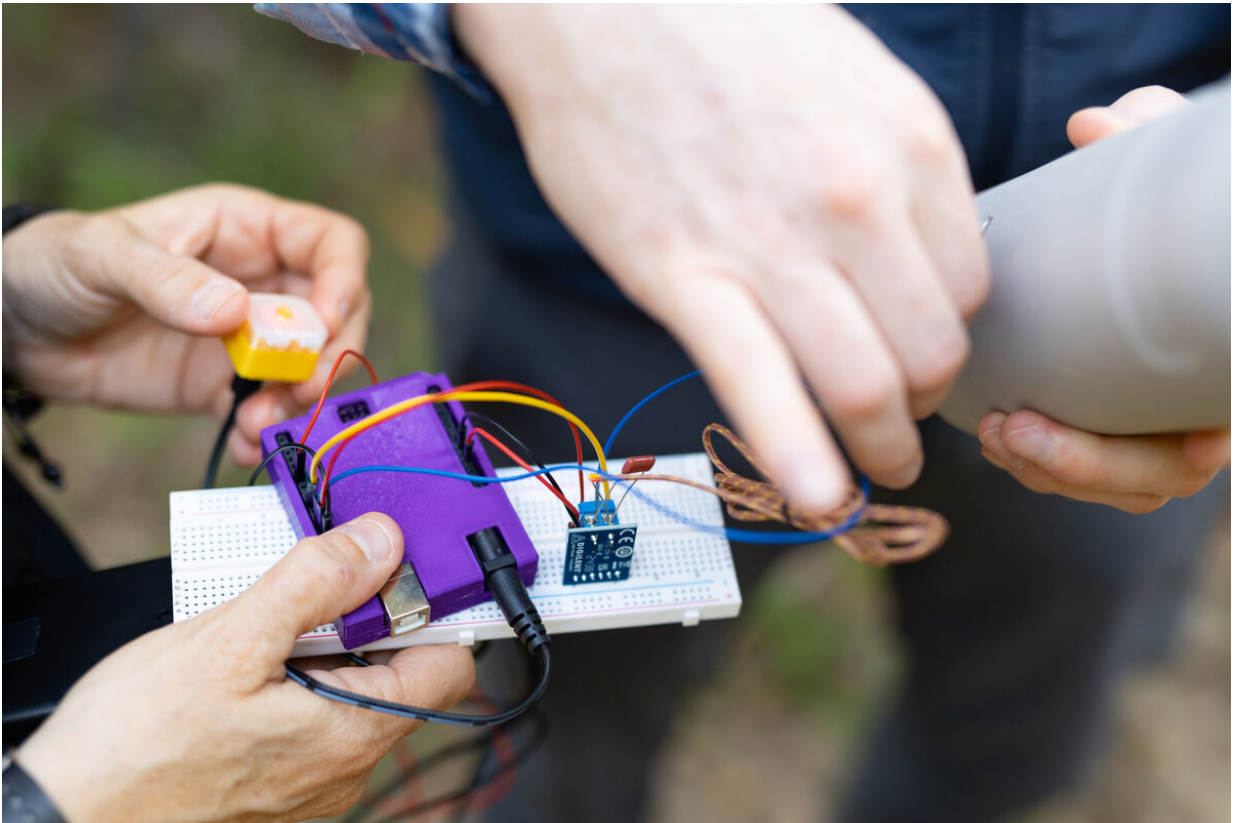


Figure C11.2: Breadboarded Sensor Module with Digital Readout — Field Data Collection



Figure C11.4: Handheld Digital Multimeter — a Common Measurement Instrument



Figure C11.5: Wind-Tunnel Data Acquisition System Computer Room

Chapter 12: Electric Motors

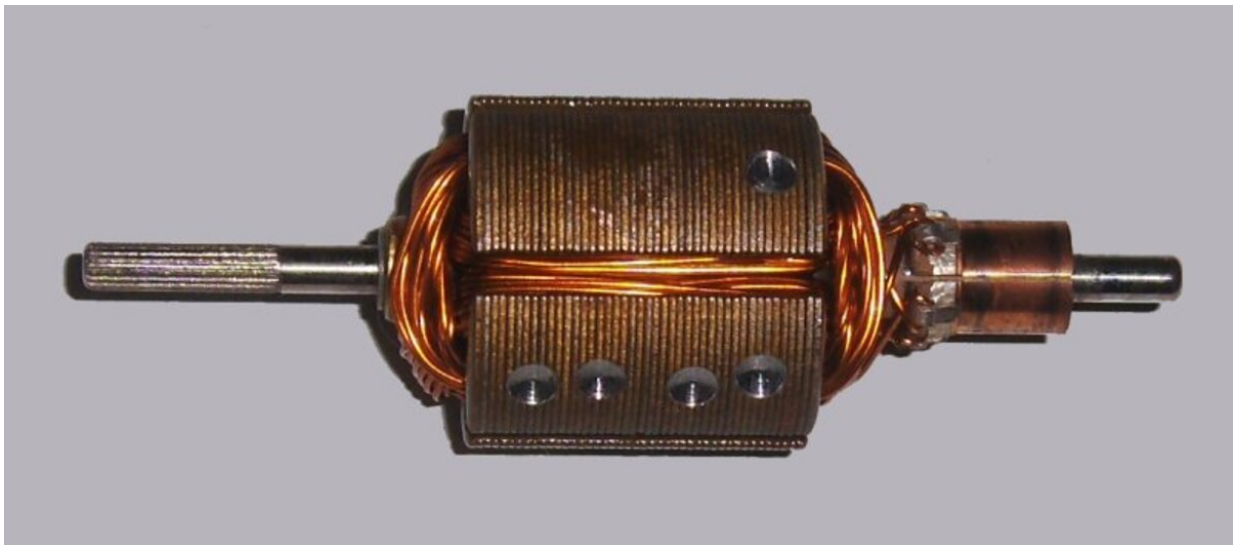


Figure C12.1: DC Motor Armature with Commutator and Windings

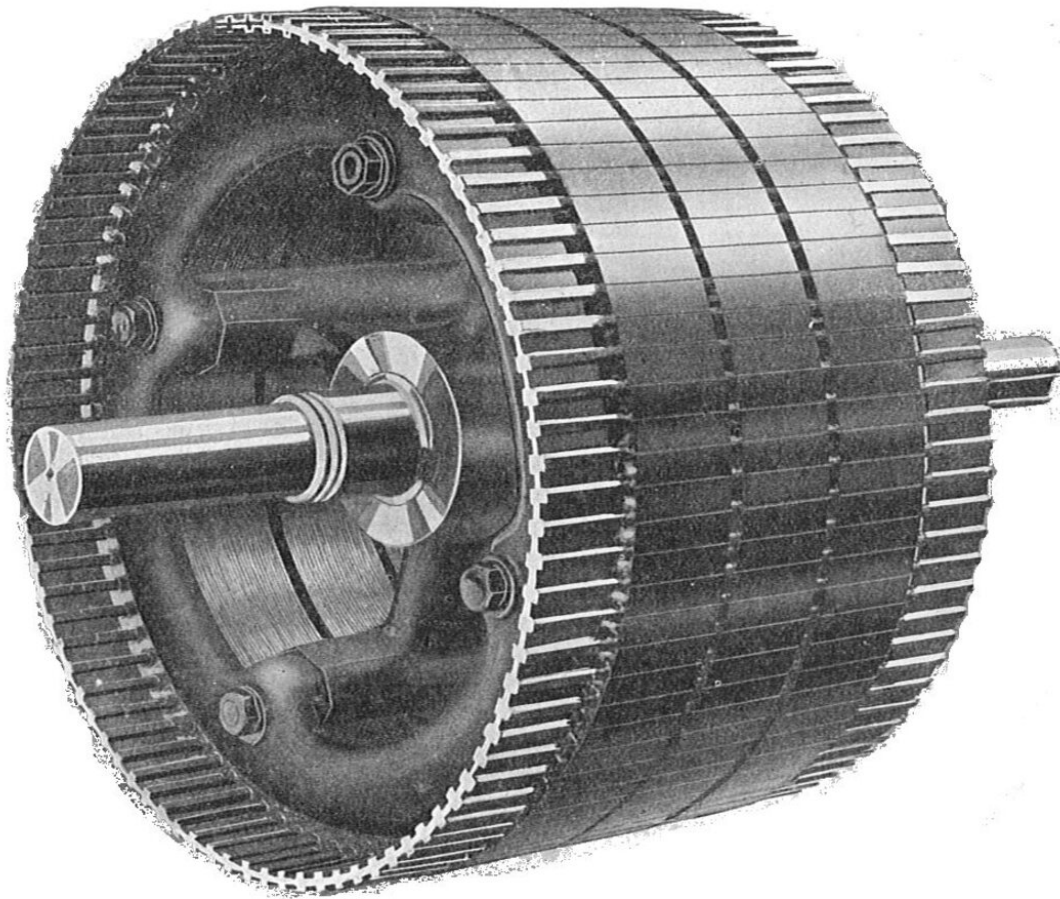


Figure C12.2: Squirrel-Cage Induction Motor Rotor (1909)

Chapter 13: Operational Amplifiers

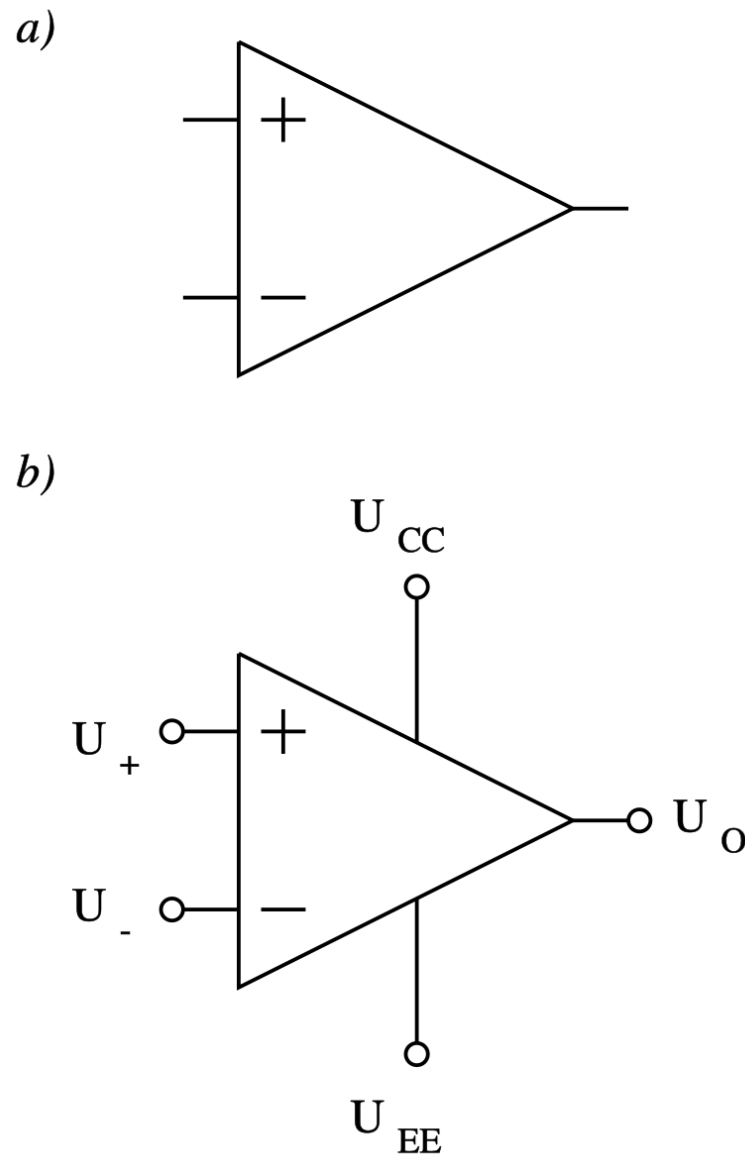


Figure C13.1: Ideal Op-Amp Symbol, With and Without Supply Rails Shown

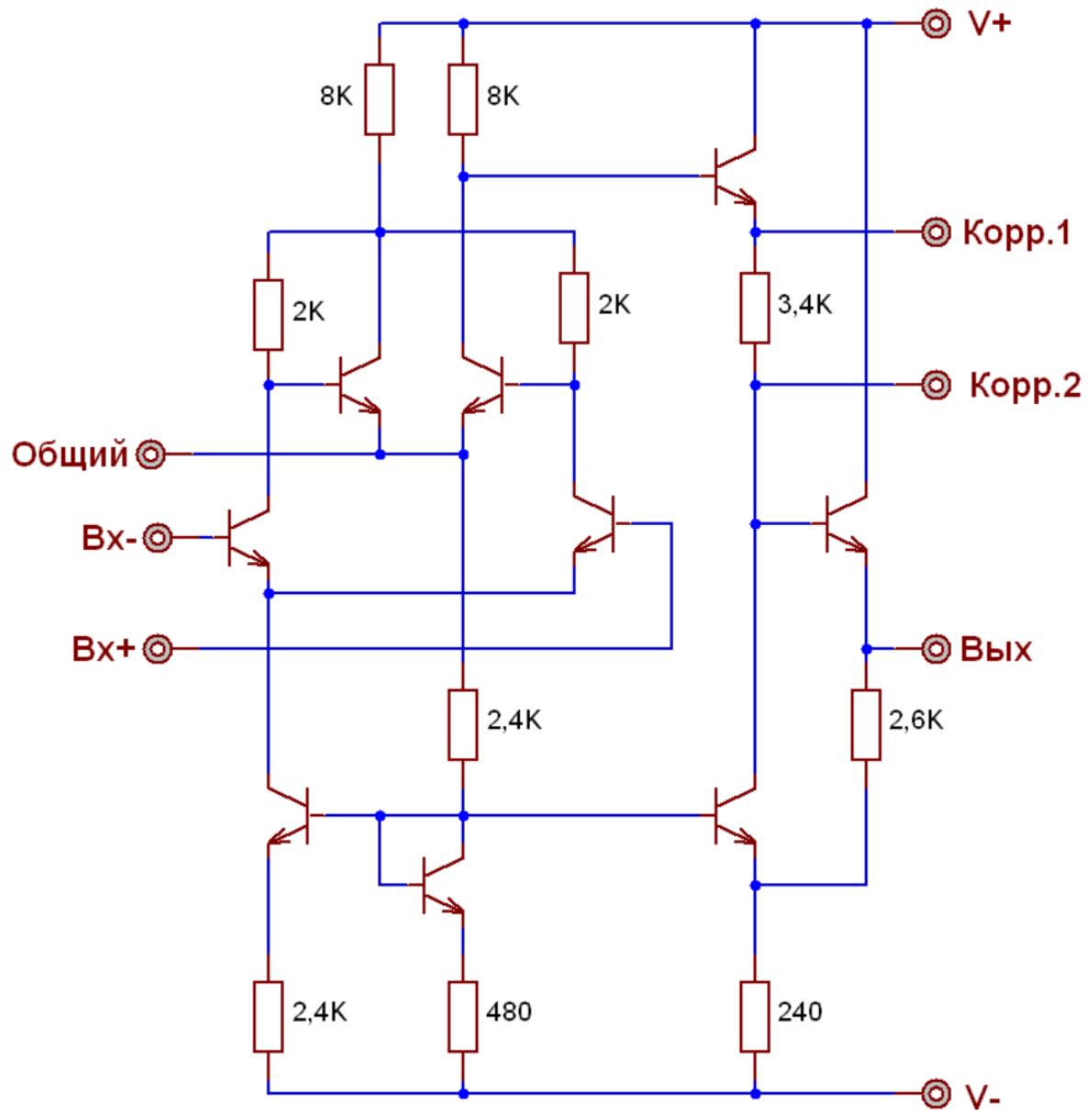


Figure C13.2: Internal Transistor-Level Schematic of the $\mu A702$, the First Monolithic Op-Amp (Bob Widlar, 1964)

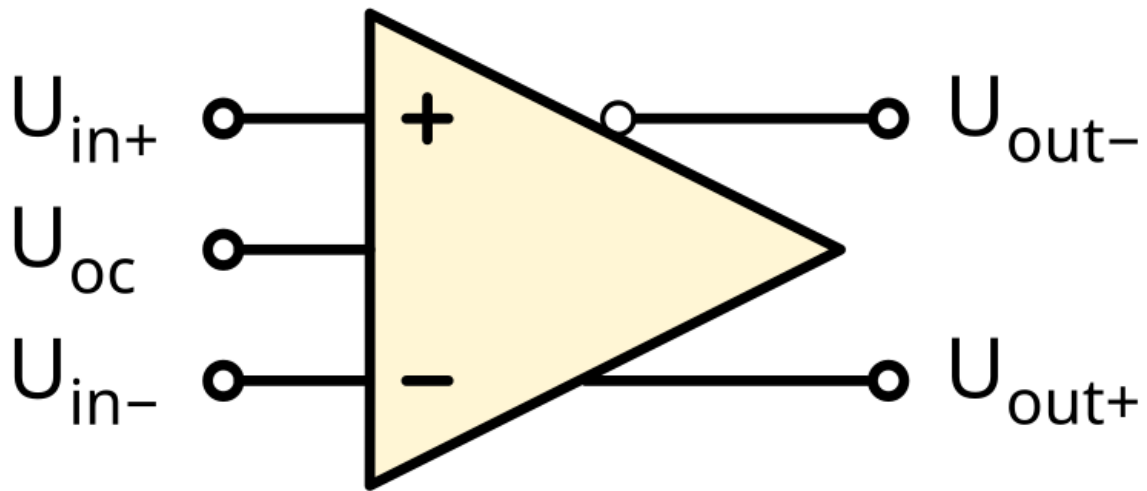


Figure C13.4: Fully-Differential Amplifier Symbol

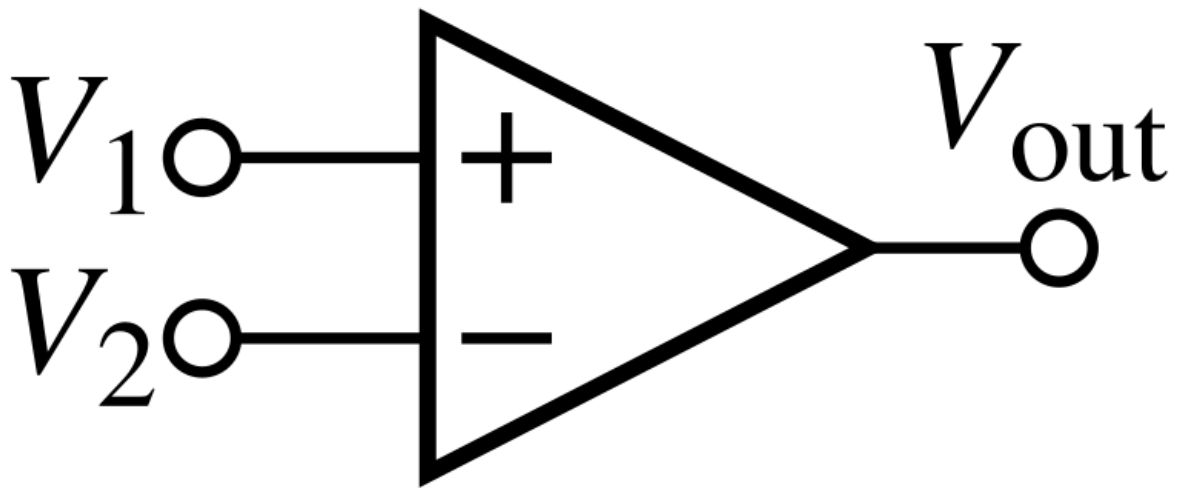


Figure C13.6: Op-Amp Used as a Comparator — Symbol and Convention

Chapter 14: National Electrical Code (NEC)



Figure C14.2: Weatherproof Electrical Device Box for Conductor Terminations



Figure C14.6: Pad-Mounted Distribution Transformer

Chapter 15: Networking

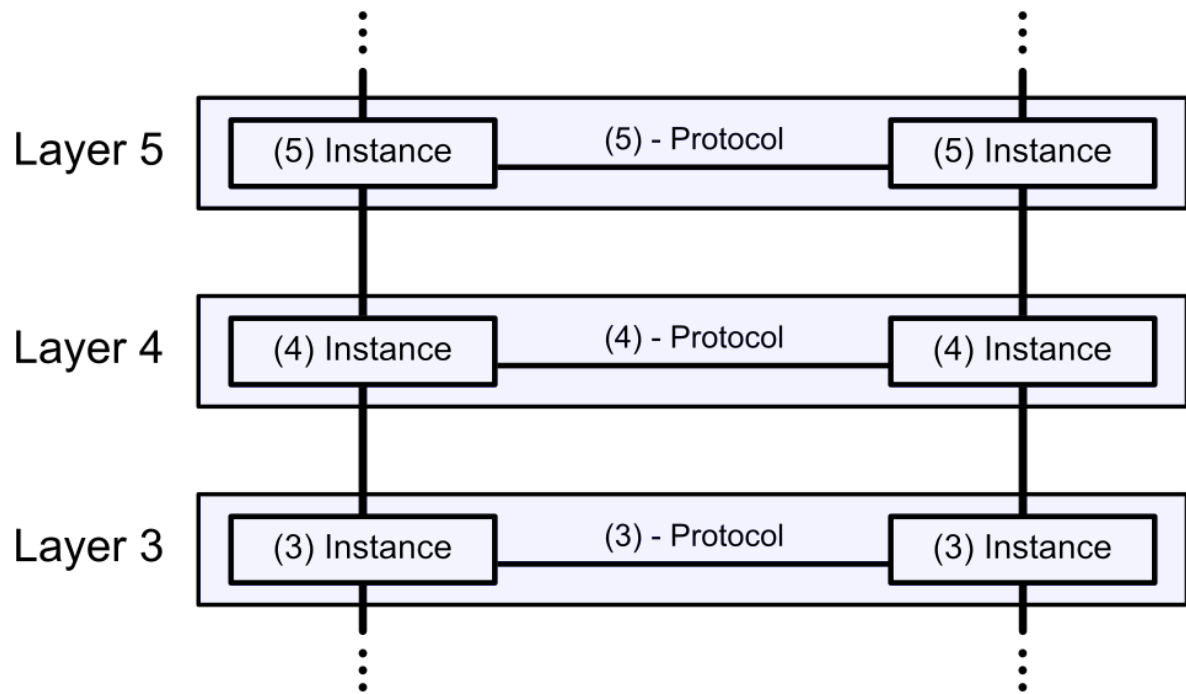


Figure C15.1: Peer-Layer Protocol Communication — the OSI Model's Layered Abstraction

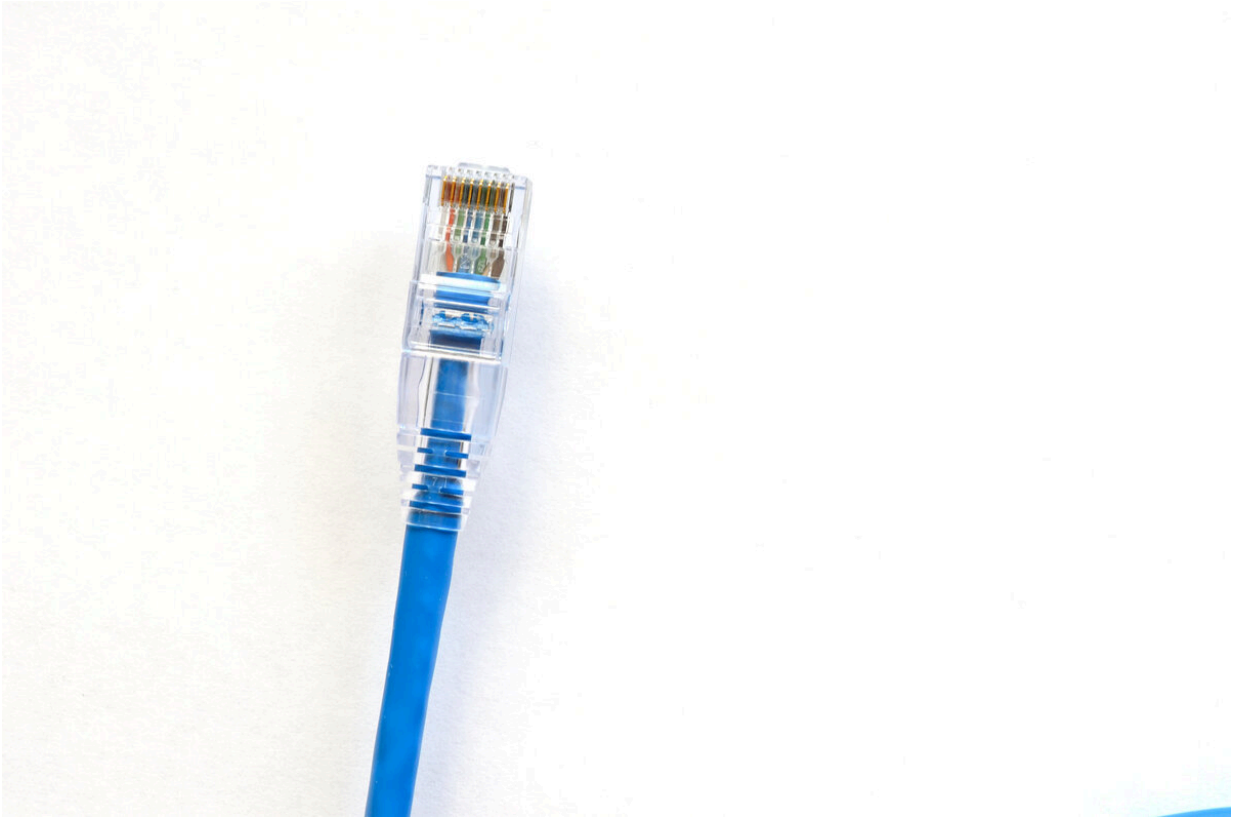


Figure C15.2: Twisted-Pair Copper Ethernet Cable



Figure C15.3: Installing Fiber-Optic Cabling for Rural Broadband (USDA Rural Development, 2024)

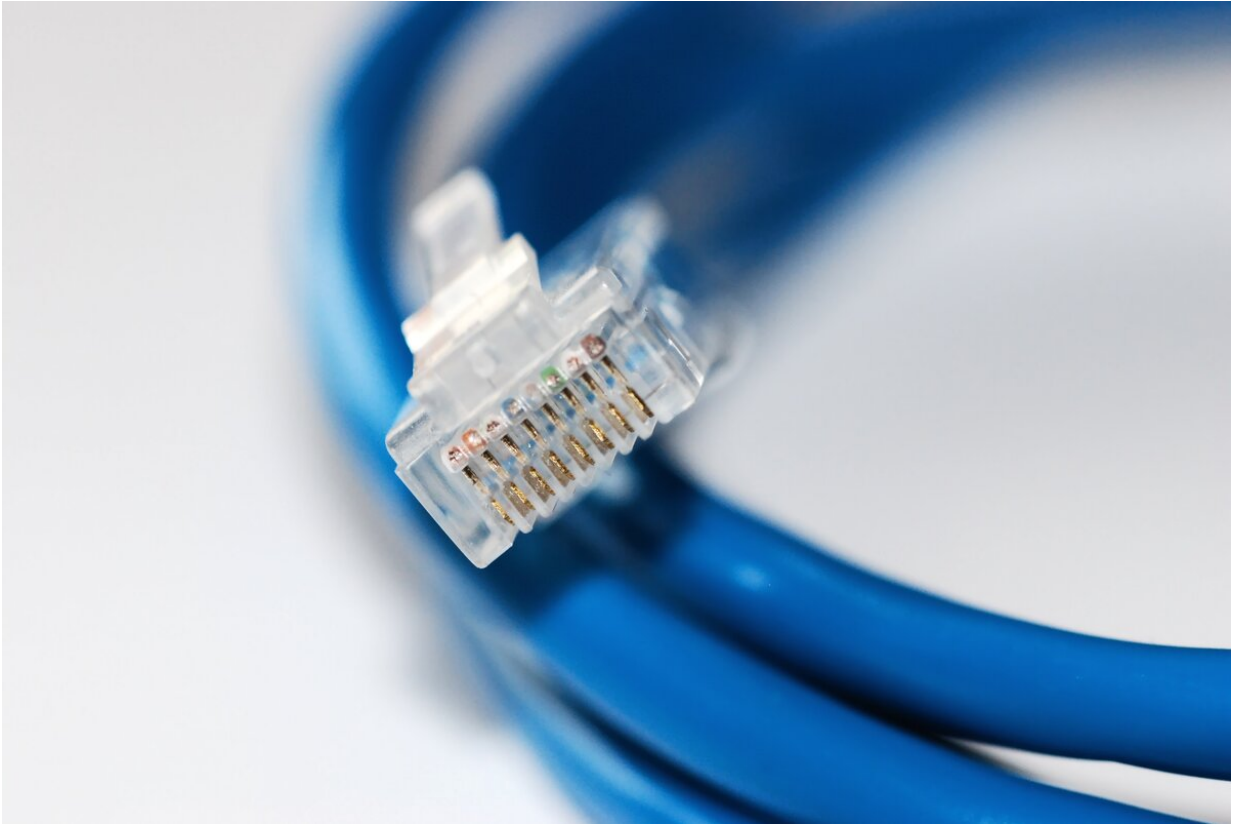


Figure C15.5: RJ45 (8P8C) Connector Used for Ethernet Twisted-Pair Cabling

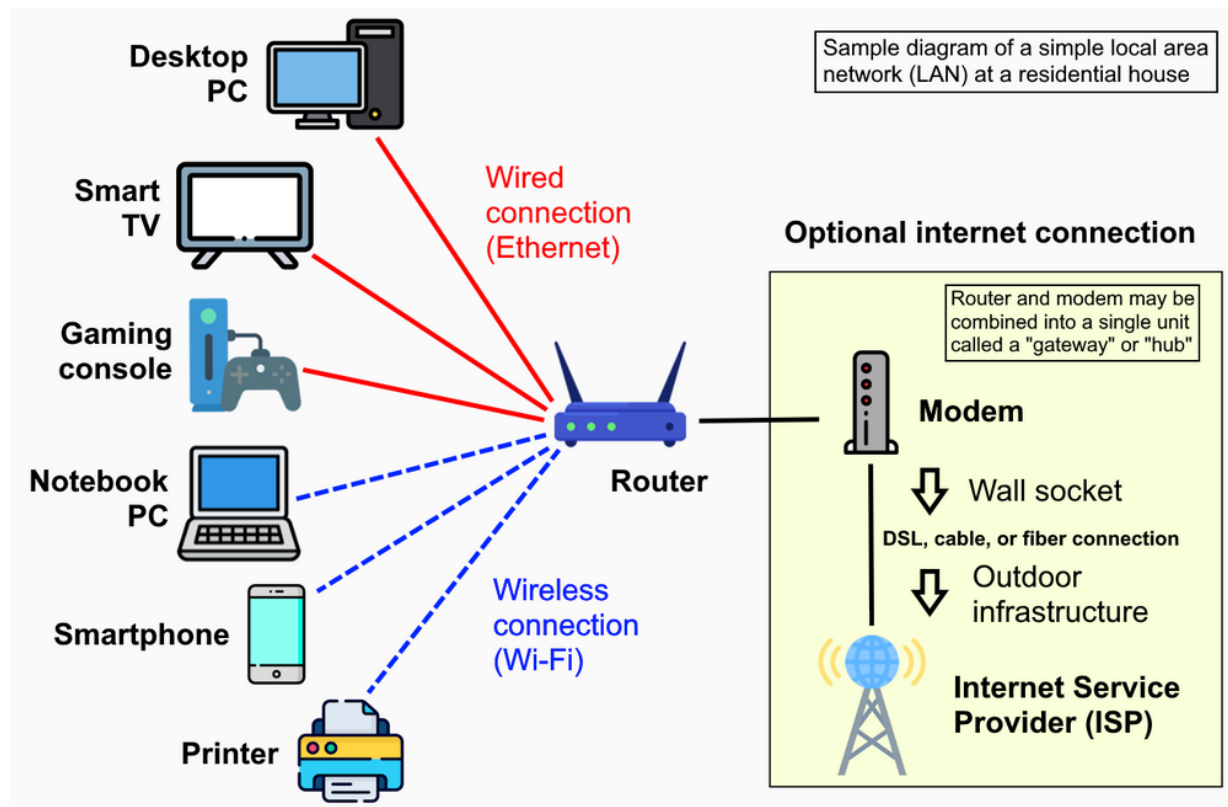


Figure C15.6: Example Home LAN / IP Network Diagram

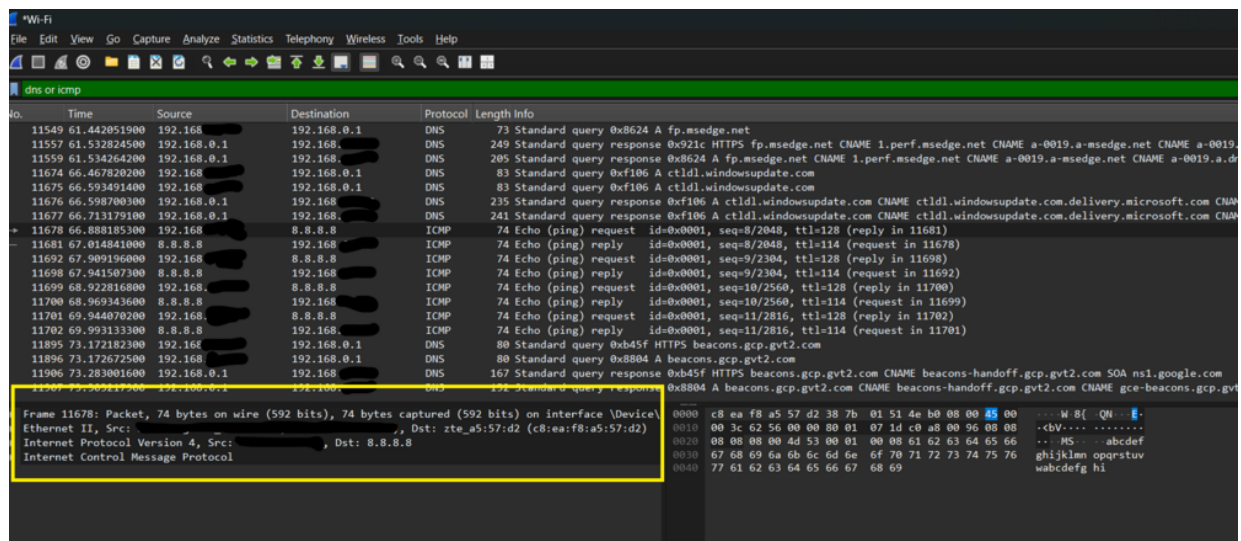


Figure C15.7: Wireshark Packet Capture — Inspecting Transport/Network-Layer Protocol Traffic



Figure C15.8: Ceiling-Mounted Wireless (Wi-Fi) Access Point



Figure C15.10: Network Cable Tester — Verifying Link Performance and Wiring Continuity

Chapter 16: Antenna Design and Principles

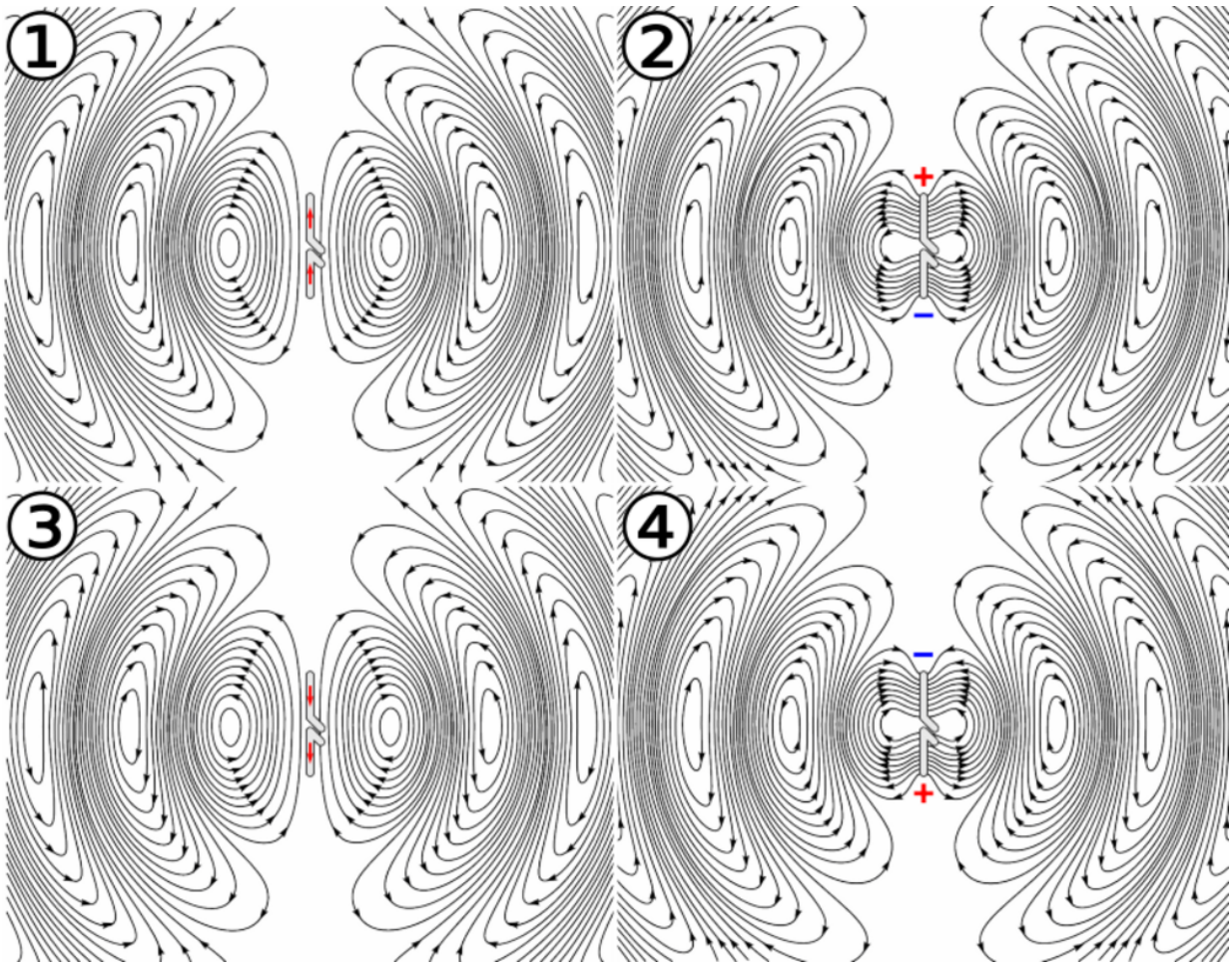


Figure C16.1: Half-Wave Dipole Antenna Field Patterns (Transmit and Receive)



Figure C16.2: UHF Ground-Plane (Quarter-Wave Whip) Antenna

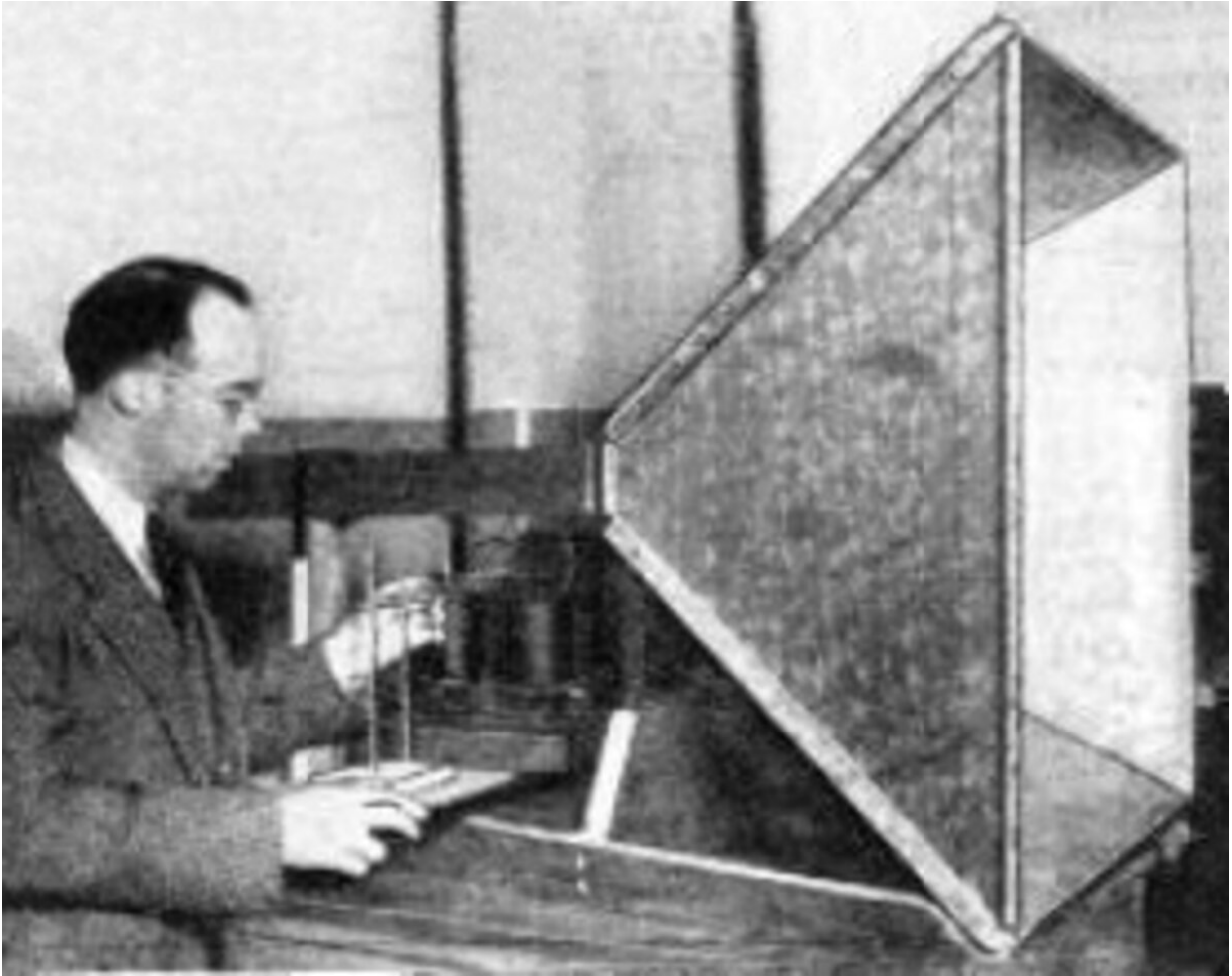


Figure C16.3: Wilmer Barrow and the First Modern Horn Antenna (1938)



Figure C16.5: AN/FPS-85 Phased-Array Radar Antenna Building, Eglin AFB

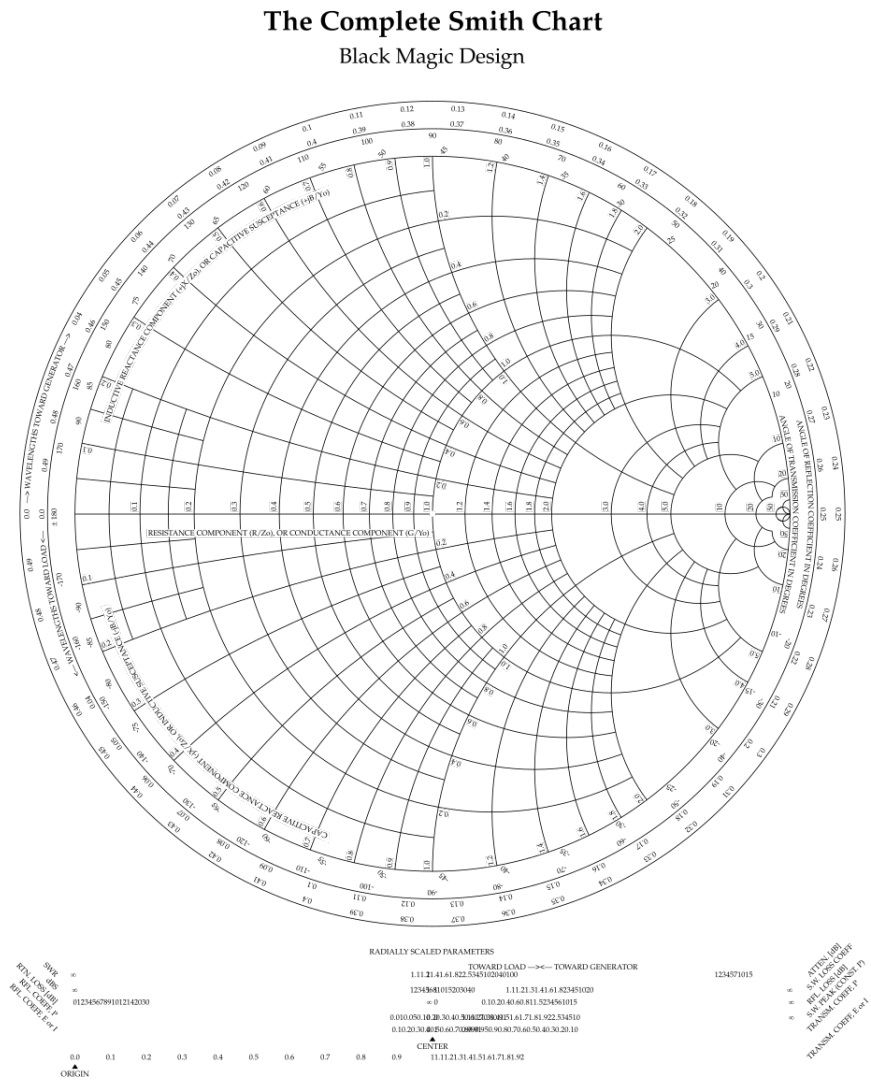


Figure C16.6: Smith Chart Used for Antenna/Transmission-Line Impedance Matching

Chapter 17: Radar Systems



Figure C17.3: PAVE PAWS Radar Operator Console — Processed Track Display



Figure C17.4: WSR-1 Weather Radar and Radiosonde Dome, North Omaha Airport



Figure C17.5: NATO E-3 AWACS — Airborne Early-Warning Radar System

Chapter 18: Optics

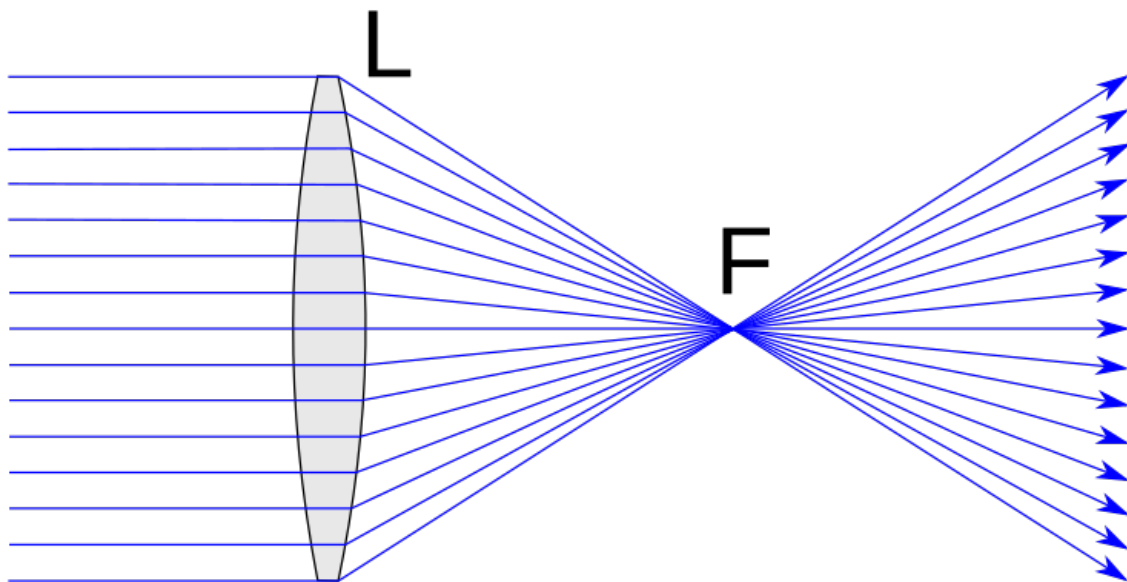


Figure C18.1: Ray Diagram Through a Perfect Convex (Converging) Lens

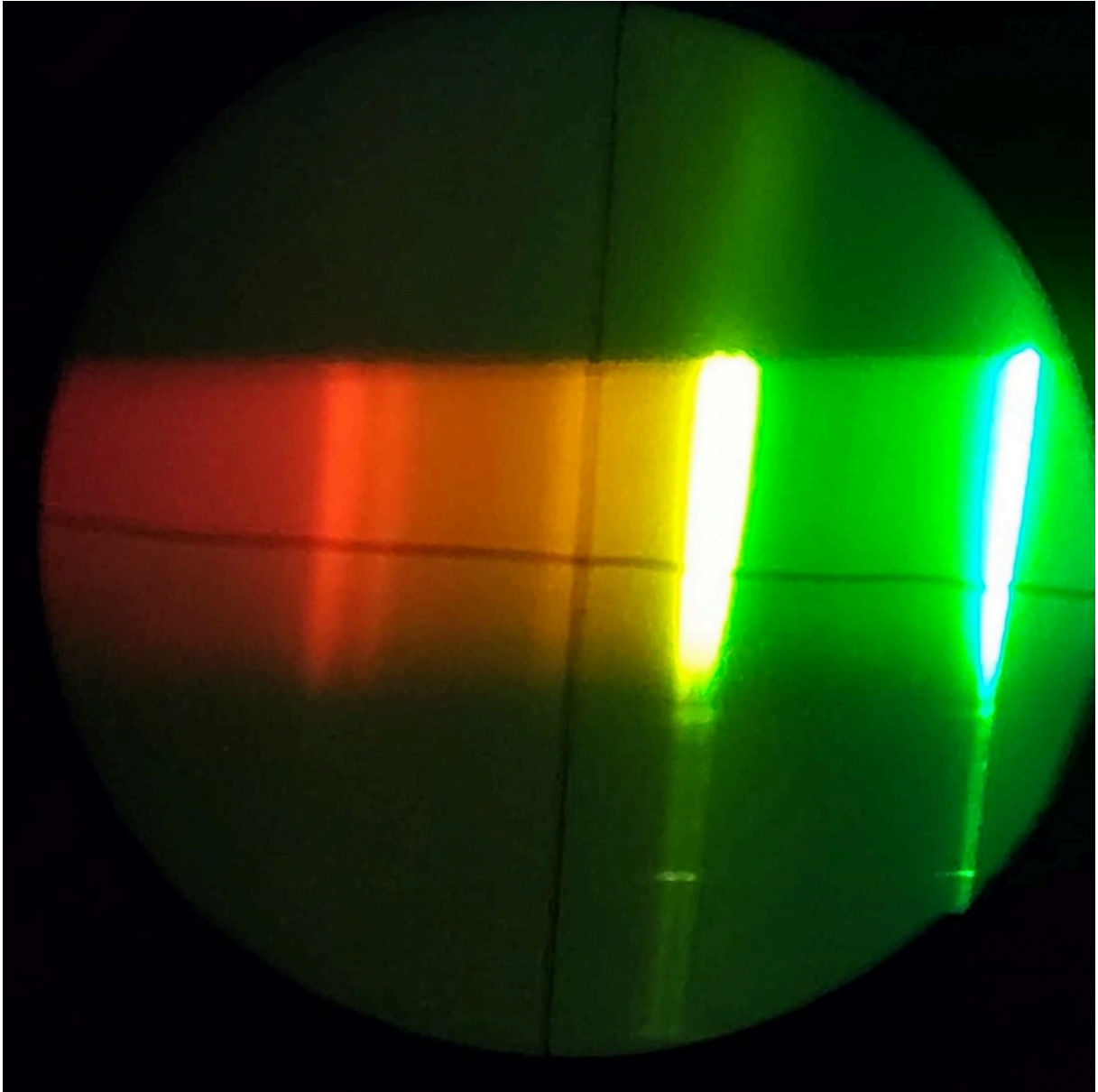


Figure C18.2: Emission Spectrum Resolved by a Diffraction Grating



Figure C18.3: NASA Researcher Aligning a Laser Device for a Laser-Powered Model Aircraft Demonstration

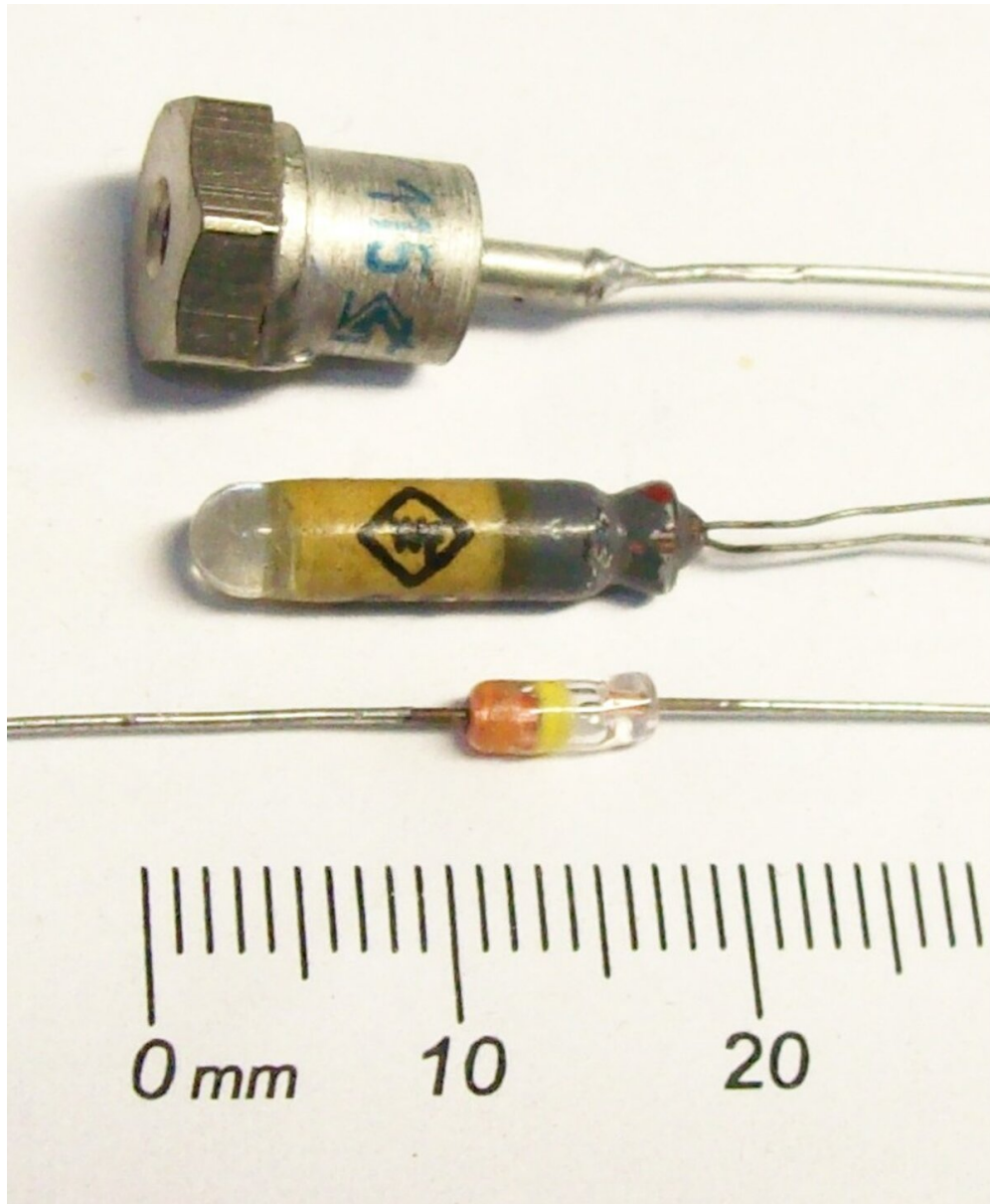


Figure C18.4: Germanium Photodiode



Figure C18.6: Hubble Space Telescope Primary Mirror — a Precision Optical System

Chapter 19: Engineering Economics



Figure C19.11: U.S. Currency — Illustrating Purchasing Power and Inflation

Appendix A: Imaginary Numbers and Phasors



Figure CA.4: Charles Proteus Steinmetz, Pioneer of the Phasor Method for AC Circuit Analysis

Appendix B: Arctangent and atan2

Appendix C: Decibels

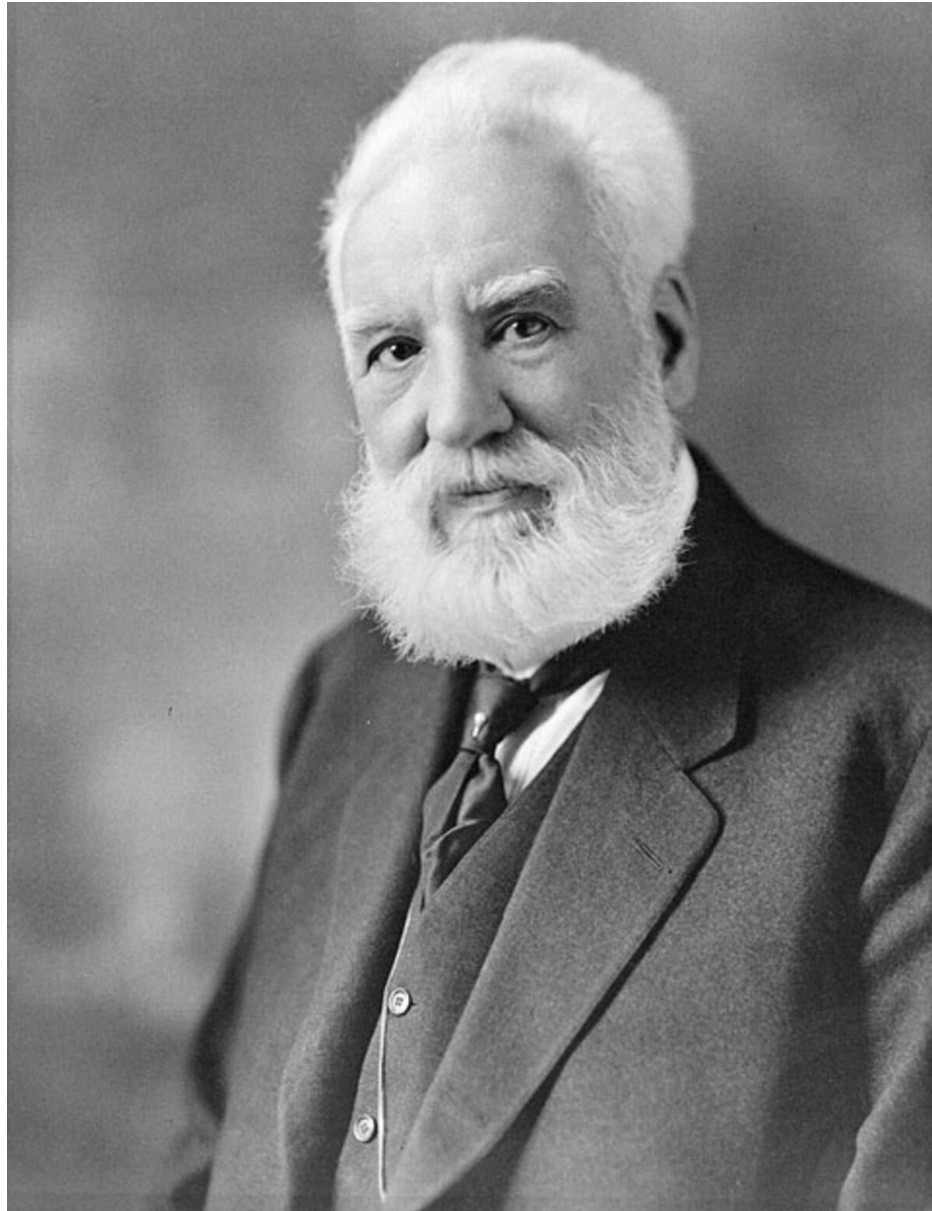


Figure CC.1: Alexander Graham Bell, Namesake of the Bel/Decibel

Appendix D: Unit Prefixes and SI Units



Figure CD.1: U.S. National Prototype Kilogram K20 — a Historical SI Base Unit Standard

Appendix E: Getting Started with Python and marimo

Appendix F: Matrix Operations

Appendix G: TCLab Control Lab